Lecture 17. scRNAseq and Gene Regulation

Michael Schatz

October 28, 2026

Applied Comparative Genomics



Project Proposal
Due Wednesday Oct 21 @ 11:59pm

Project Proposal

Assignment Date: Monday October 12, 2026
Due Date: Wednesday, October 21 2026 @ 11:59pm

Review the Project Ideas page

Work solo or form a team for your class project of no more than 3 people.

The proposal should have the following components:

- Name of your team
- List of team members and email addresses
- Short title for your proposal
- 1 to 2 paragraph description of what you hope to do and how you will do it
- References to 3 to 4 relevant papers
- References/URLs to datasets that you will be studying (Note you can also use simulated data)
- Please add a note if you need me to sponsor you for a MARCC/skipjack account (high RAM, GPUs, many cores, etc)
- Please add a note if you need me to sponsor you for an AnVIL cloud billing account (dynamic resources)

Submit the proposal as a 1 to 2 page PDF on GradeScope (each team should submit one proposal and tag all people in the team). After submitting your proposal, I will provide feedback. If necessary, we can schedule a time to discuss your proposal, especially to ensure you have access to the data that you need. The sooner that you submit your proposal, the sooner we can schedule the meeting.

Later, you will present your project in class during the last week of class. You will also submit a written report (5-7 pages) of your project, formatting as a Bioinformatics article (Intro, Methods, Results, Discussion, References).

Please use Piazza to coordinate proposal plans!

Assignment 4: RNA-seq and Machine Learning

Assignment Date: Monday, October 26, 2026
Due Date: Monday, November 2 @ 11:59pm

Assignment Overview

In this assignment you will explore a couple of key aspects of RNA-seq and introduce the key concepts of machine learning. For this assignment, we will provide a Jupyter notebook with code for you to use and complete your assignment in.

To use the Jupyter notebook, create a new mamba environment for assignment 4:

```
mamba create -n asn4 scikit-learn pytorch matplotlib jupyter umap-learn seaborn pandas numpy
```

Make sure to download the training data for the assignment from Dropbox. Place this in the `data_files` directory and make sure to uncompress (`gunzip`) both the training and testing files.

As a reminder, any questions about the assignment should be posted to Piazza

See the notebook here: Assignment4.ipynb

Packaging

The solutions to the above questions should be submitted as a single PDF document that includes your name, email address, and all relevant figures (as needed). If you use ChatGPT for any of the code, also record the prompts used. Submit your solutions by uploading the PDF to GradeScope, and remember to select where in your submission each question/subquestion is.

If you submit after this time, you will use your late days. Remember, you are only allowed 4 late days for the entire semester!

Resources

- Jupyter notebooks: https://jupyter.org/
- scikit-learn: https://scikit-learn.org/stable/
- pytorch: https://pytorch.org/
- umap-learn: https://umap-learn.readthedocs.io/en/latest/basic_usage.html

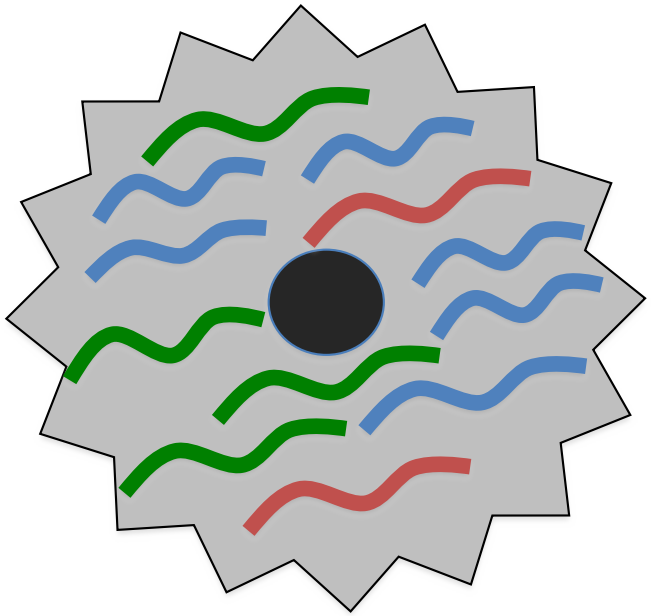
Annotation Summary

- Three major approaches to annotate a genome
  1. Alignment:
     - Does this sequence align to any other sequences of known function?
     - Great for projecting knowledge from one species to another
  2. Prediction:
     - Does this sequence statistically resemble other known sequences?
     - Potentially most flexible but dependent on good training data
  3. Experimental:
     - Lets test to see if it is transcribed/methylated/bound/etc
     - Strongest but expensive and context dependent



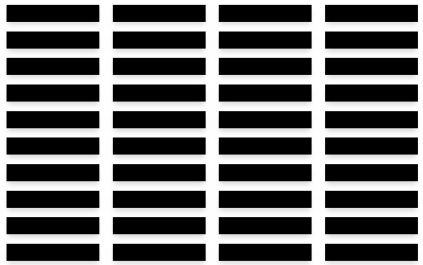
- Many great resources available
  - Learn to love the literature and the databases
  - Standard formats let you rapidly query and cross reference
  - Google is your number one resource ☺

RNA-seq Overview



Sequencing

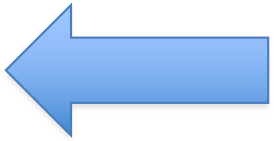




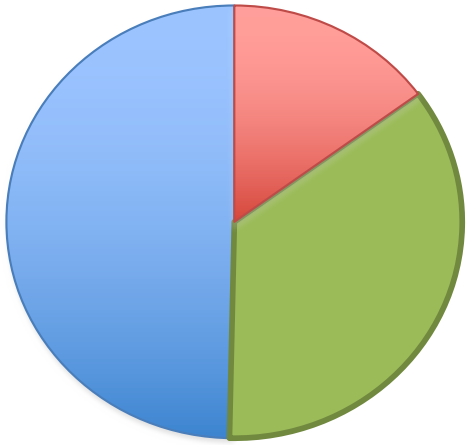


Mapping
& Assembly

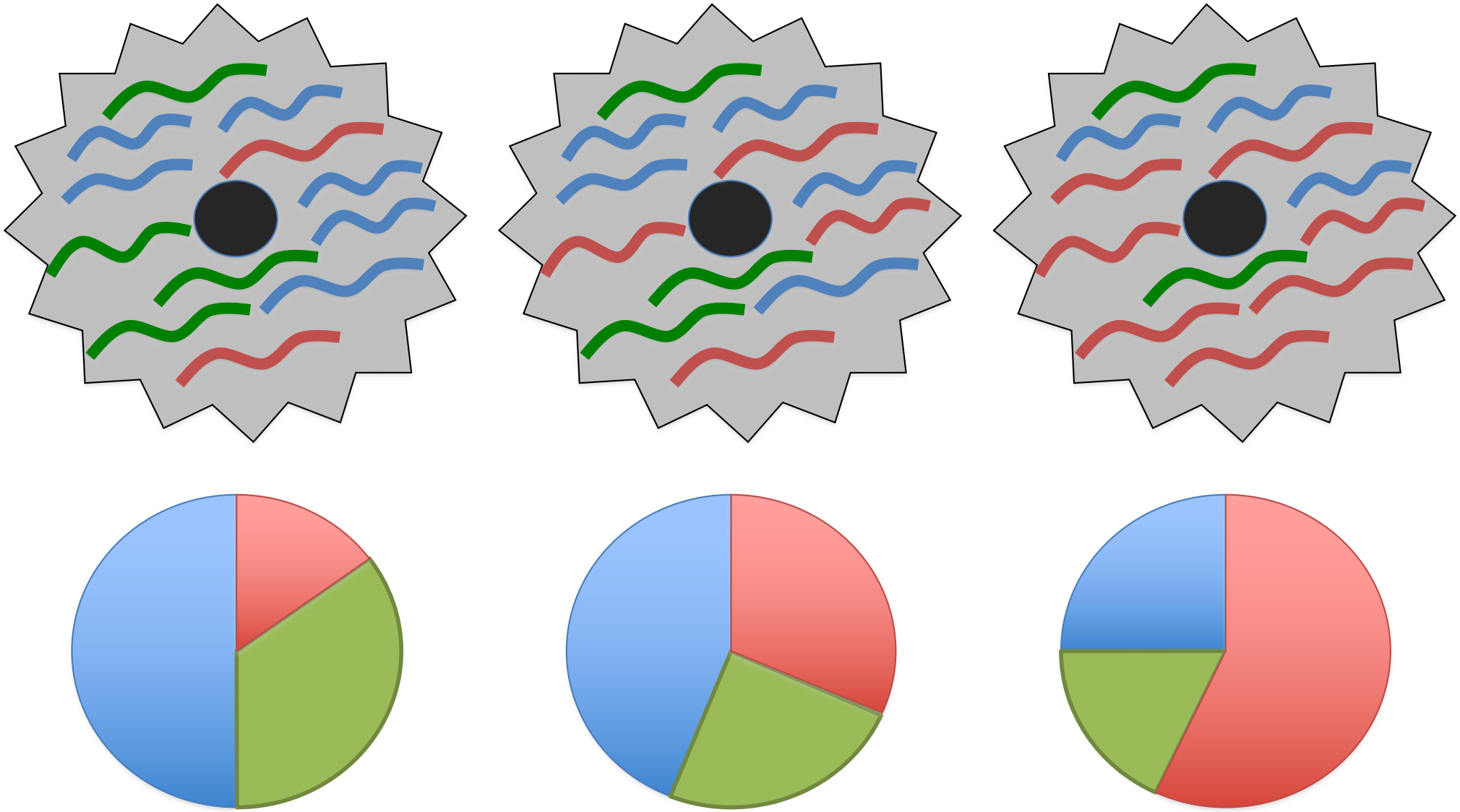




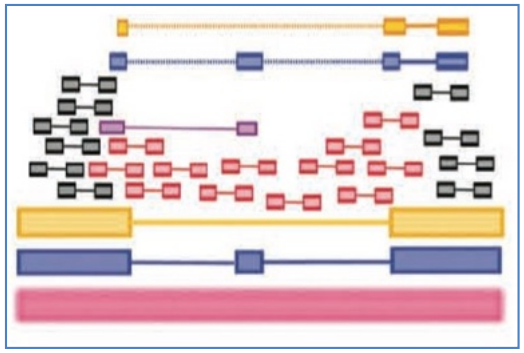
Quantification



RNA-seq Overview



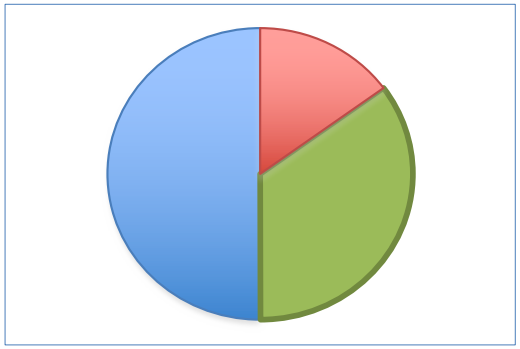
RNA-seq Challenges



Challenge 1: Eukaryotic genes are spliced

Solution: Use a spliced aligner, and assemble isoforms

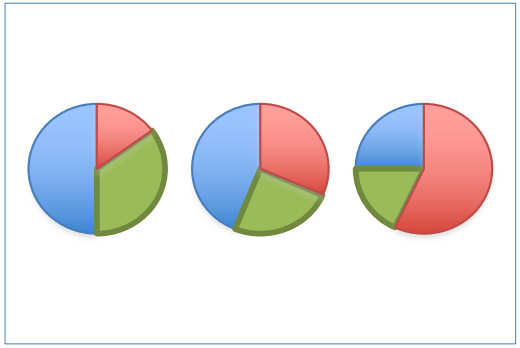
TopHat: discovering spliced junctions with RNA-Seq.
Trapnell et al (2009) *Bioinformatics*. 25:0 1105-1111



Challenge 2: Read Count != Transcript abundance

Solution: Infer underlying abundances (e.g. TPM)

Transcript assembly and quantification by RNA-seq
Trapnell et al (2010) *Nat. Biotech.* 25(5): 511-515



Challenge 3: Transcript abundances are stochastic

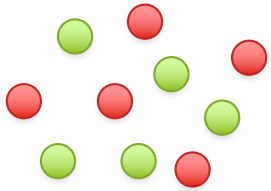
Solution: Replicates, replicates, and more replicates

RNA-seq differential expression studies: more sequence or more replication?
Liu et al (2013) *Bioinformatics*. doi:10.1093/bioinformatics/btt688

Population Heterogeneity

Red cells express twice the abundance of “brain” genes compared to green cells

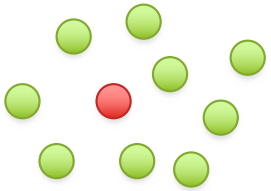
Experiment 1: 50/50



Compared to a control sample of pure green cells, this sample will show:

50% 2x + 50% 1x
= 1.5x over expression of brain genes

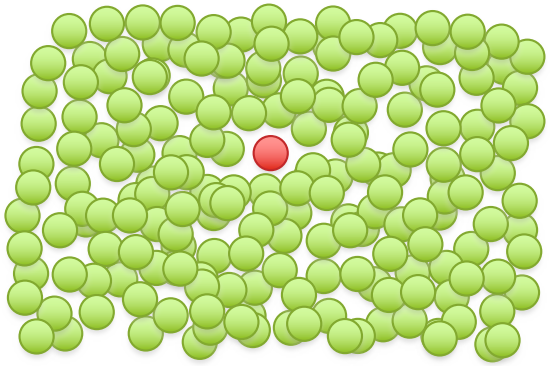
Experiment 2: 1/10



Compared to a control sample of pure green cells, this sample will show:

10% 2x + 90% 1x
= 1.1x over expression of brain genes

Experiment 3: 1/1000



Compared to a control sample of pure green cells, this sample will show:

0.1% 2x + 99.1% 1x
= 1.001x over expression of brain genes

The limitations of averages

	Drug A	Drug B
Overall Response	78% (273/350)	**83% (289/350)**

The limitations of averages

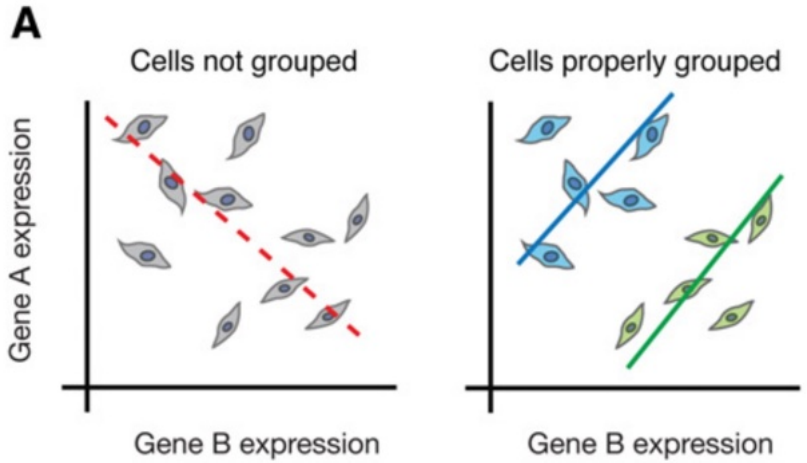
	Drug A	Drug B
Overall Response	78% (273/350)	**83% (289/350)**
Male Response	**93% (81/87)**	87% (234/270)
Female Response	**73% (192/263)**	69% (55/80)

What??? How can the better performing drug depend on if you examine the overall response or separately examine by gender

Example of Simpson's paradox:
Trend of the overall average may reverse the trends of each constituent group

In this example, the "lurking" variable (or confounding variable) is the severity of the case (represented by the doctors' treatment decision trend of favoring B for less severe cases), which was not previously known to be important until its effects were included. (Based on real analysis of kidney stone treatments)

The paradox of averages



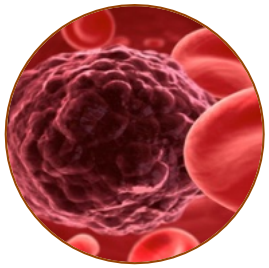
What??? How can the better performing drug depend on if you examine the overall response or separately examine by gender

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(Trapnell, 2015, Genome Research)

Sources of (Genomic) Heterogeneity



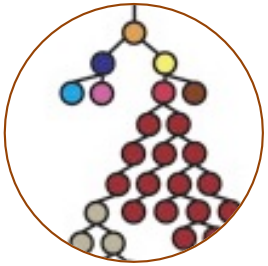
Circulating tumor cells



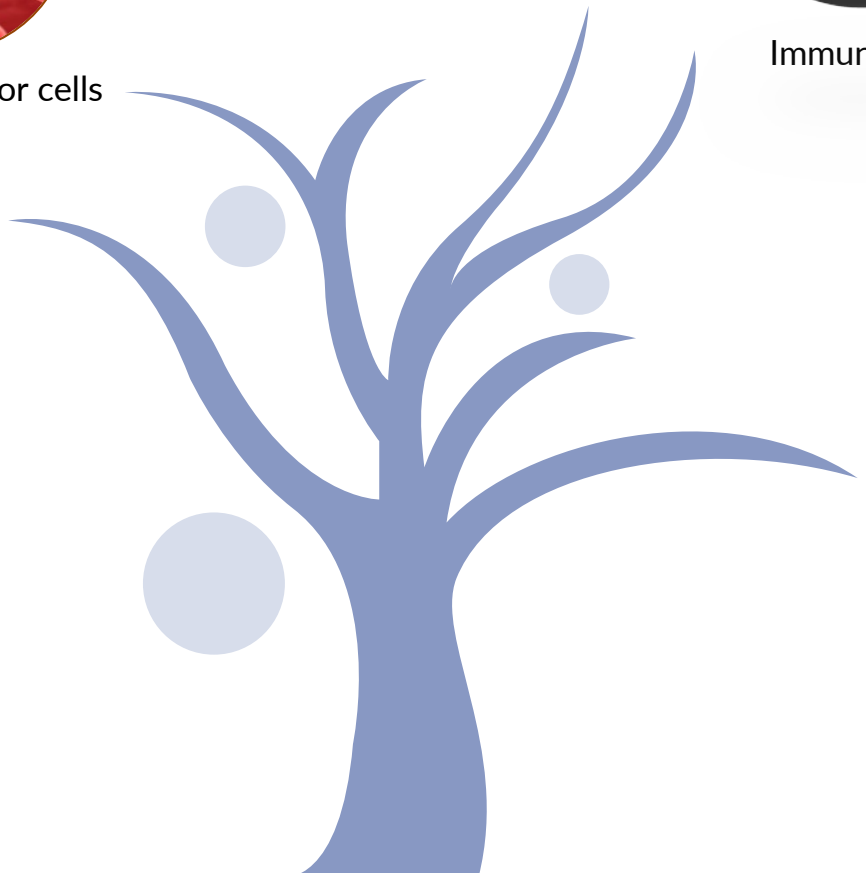
Neuronal mosaicism

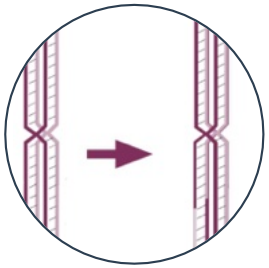


Immune Cells



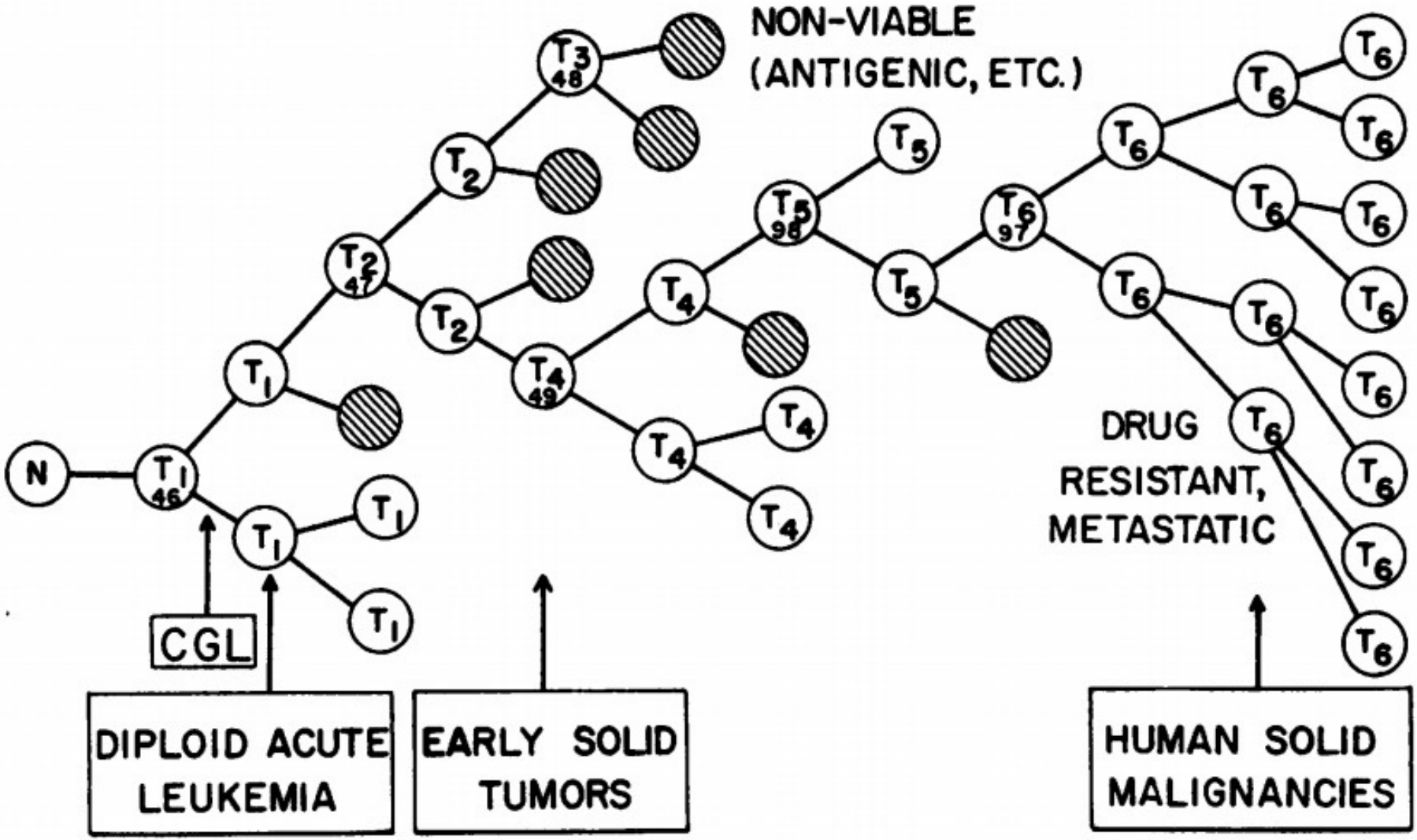
Tumor evolution





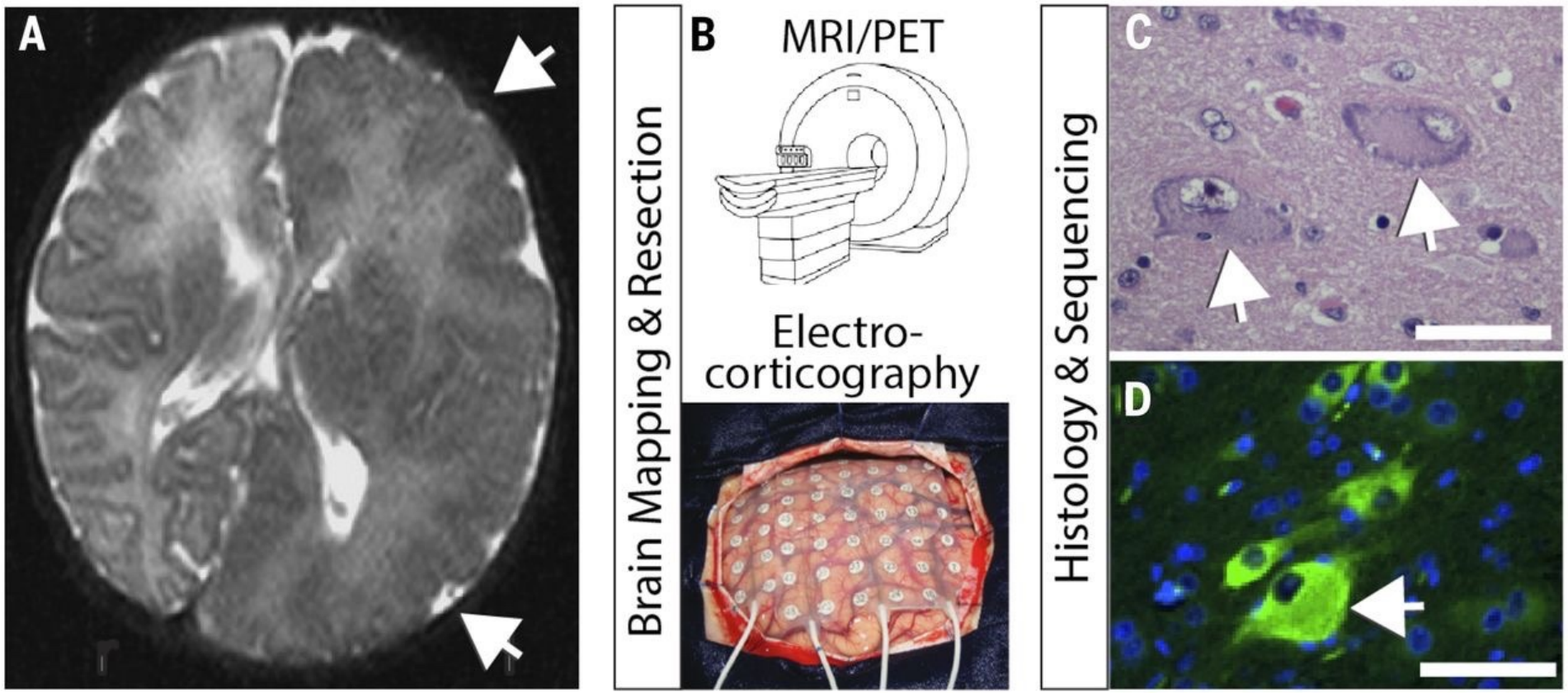
Recombination/crossover
in germ cells

Tumor Evolution



The Clonal Evolution of Tumor Cell Populations

Peter C. Nowell (1976) *Science.* 194(4260):23-28 DOI: 10.1126/science.959840

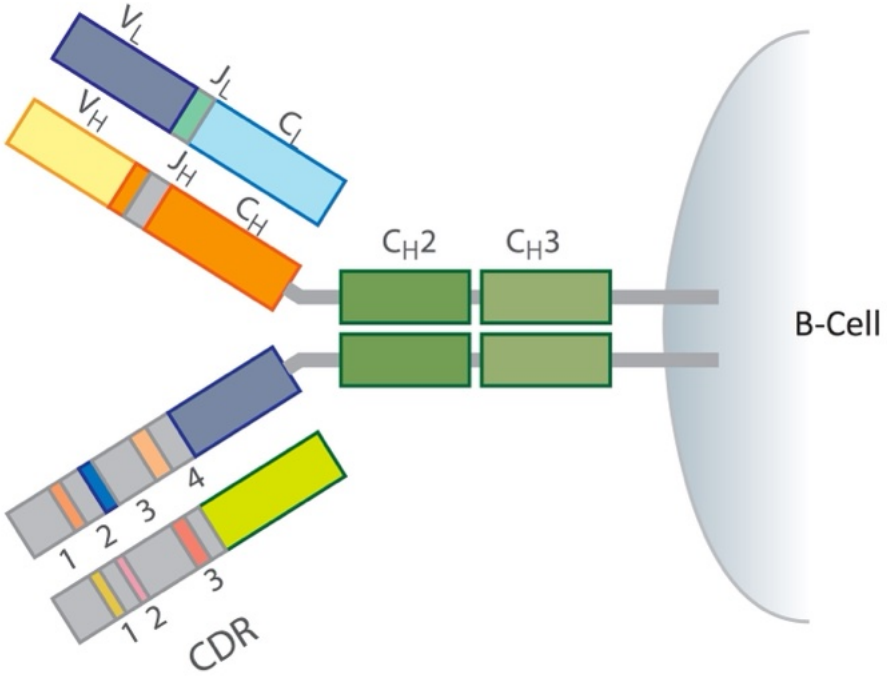


An example of brain somatic mosaicism that leads to a focal overgrowth condition.

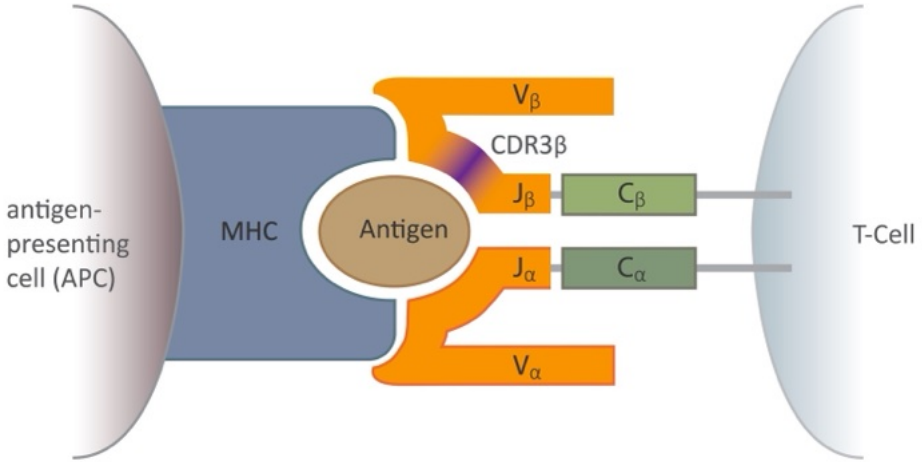
(**A**) Axial brain MRI of focal overgrowth from a 2-month-old child with intractable epilepsy and intellectual disability. (**B**) Brain mapping using high-resolution MRI is followed by surgical resection of diseased brain tissue. (**C**) Histological analysis with hematoxylin/eosin showing characteristic balloon cells consisting of large nuclei, distinct nucleoli, and glassy eosinophilic cytoplasm. (**D**) After surgery, the patient showed clinical improvement.

Intersection of diverse neuronal genomes and neuropsychiatric disease: The Brain Somatic Mosaicism Network. McConnell et al (2017) Science. doi: 10.1126/science.aal1641

Immunology



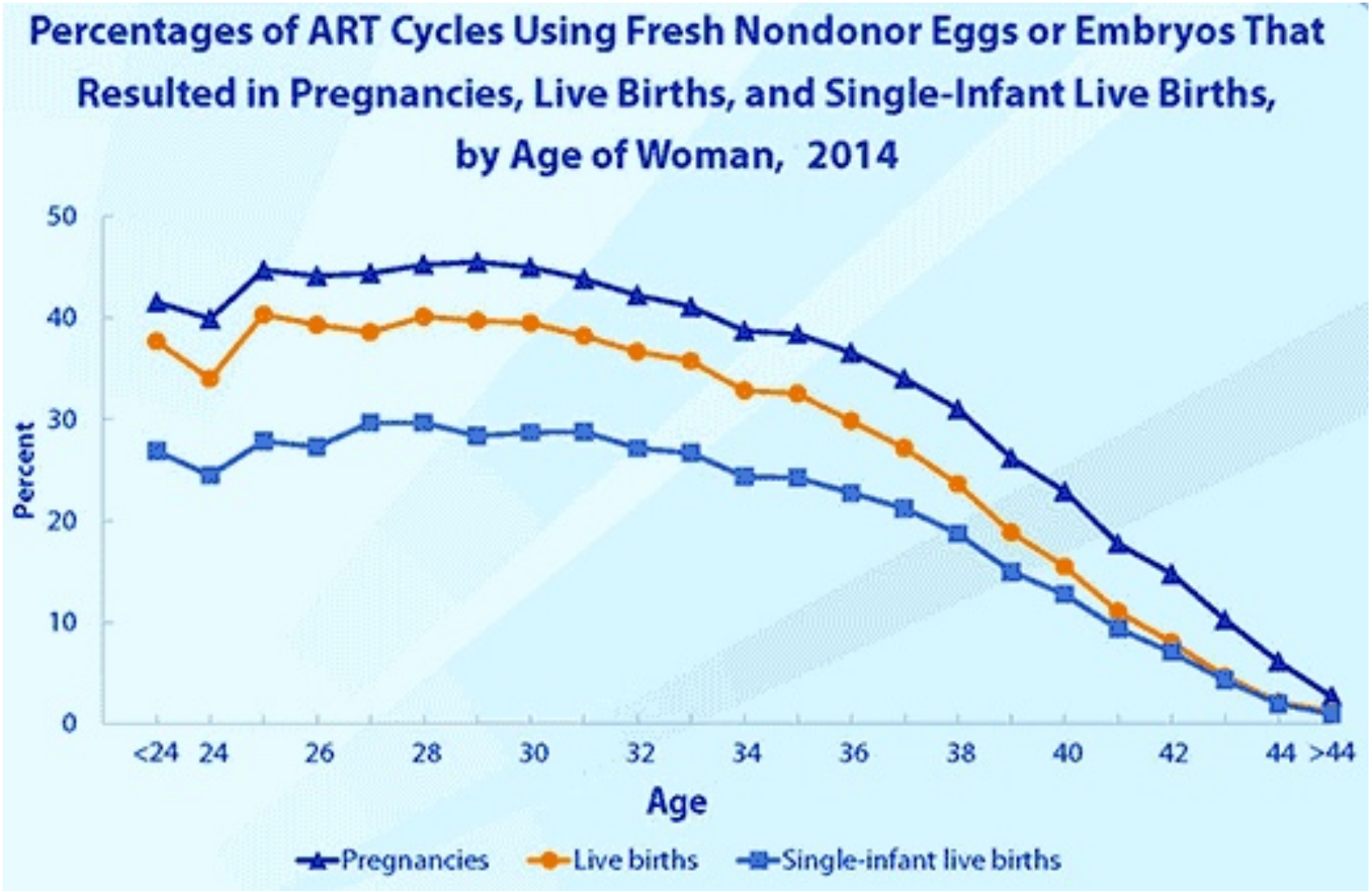
- Massive diversity rivaled only by germ cells
- Somatic recombination

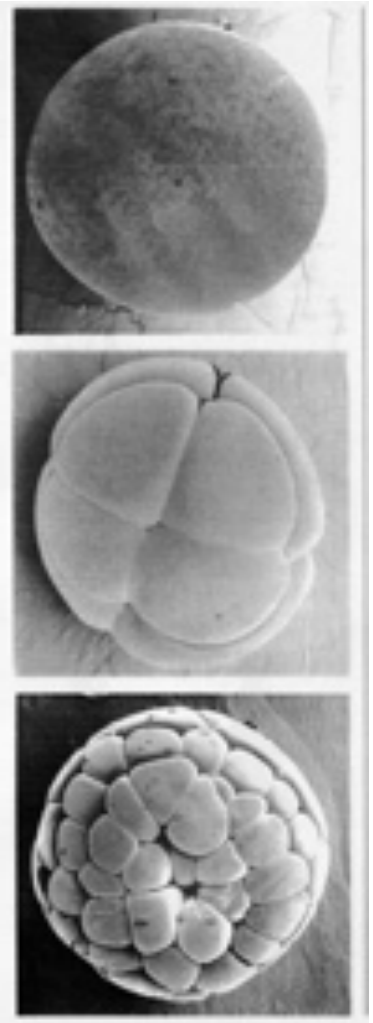


- B cells – antibody generation
- T cells – antigen response

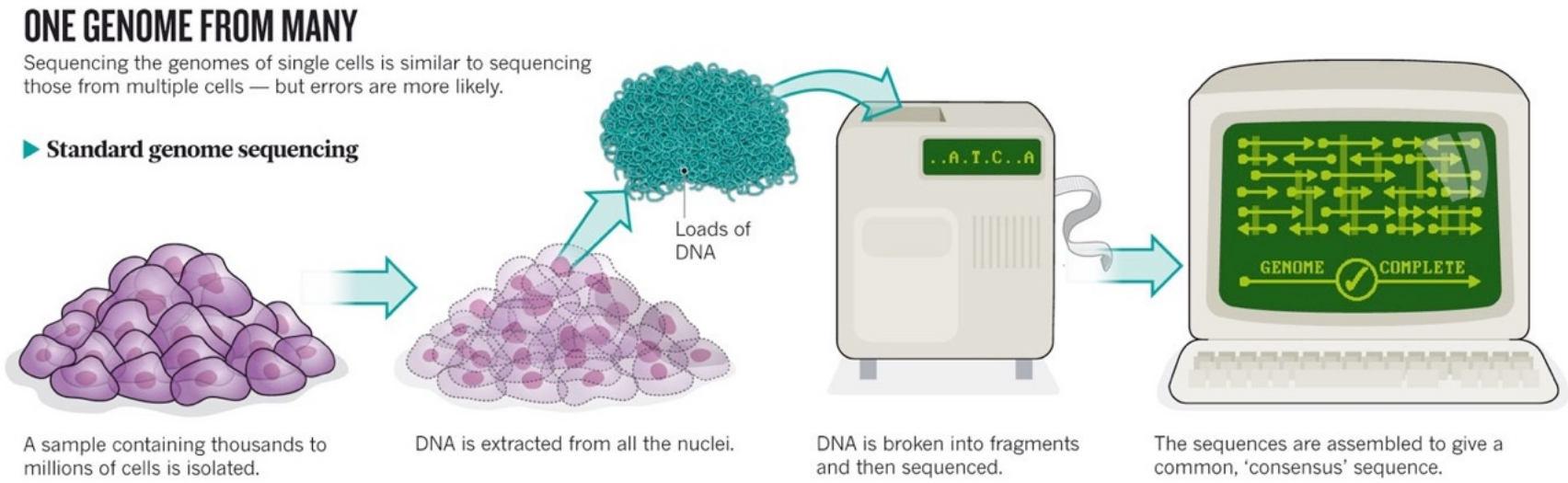
Single cell research. Illumina.

In-vitro Fertilization



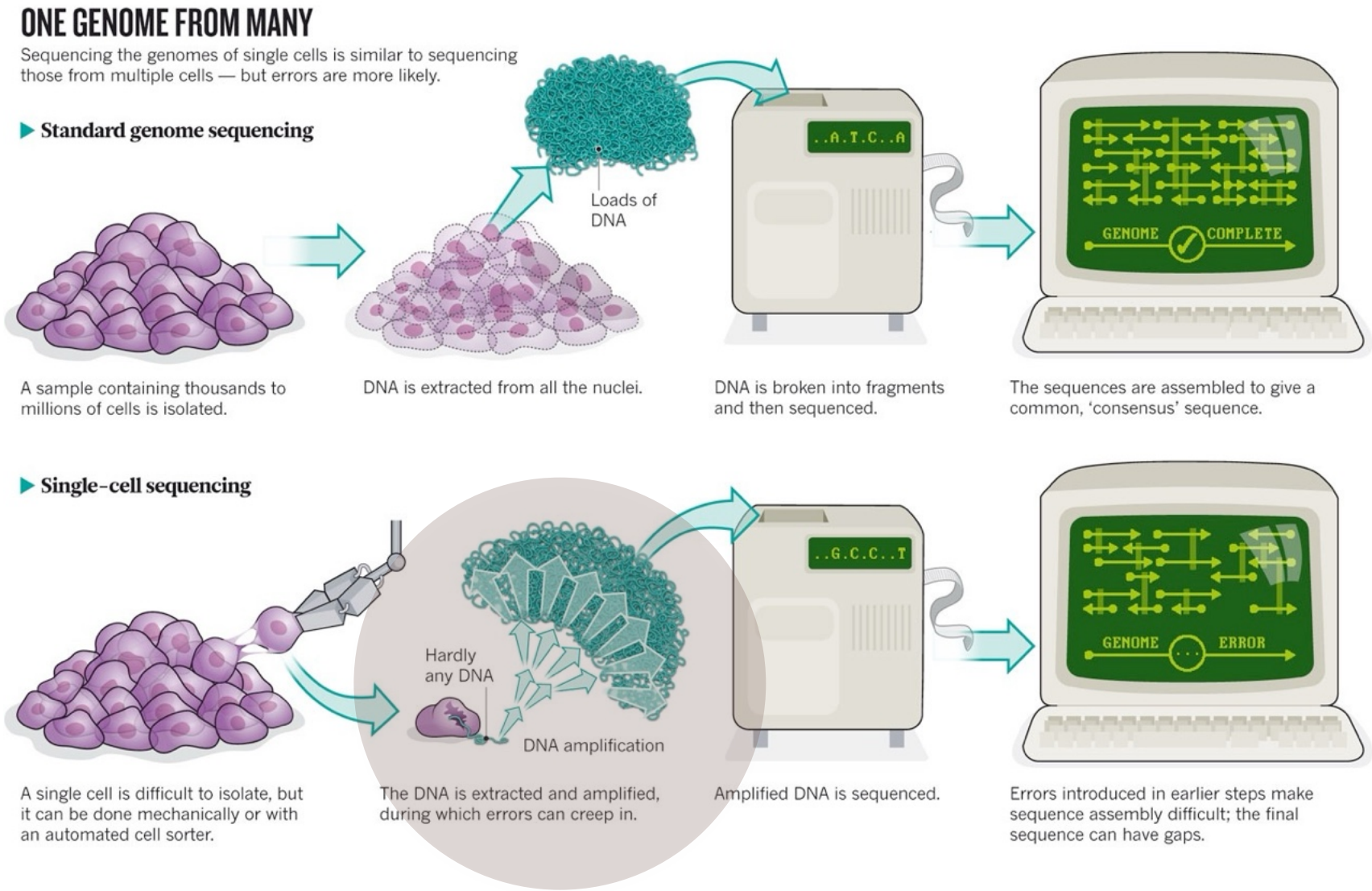


Single-cell vs. bulk sequencing



Brian Owens, Nature News 2012

Single-cell vs. bulk sequencing

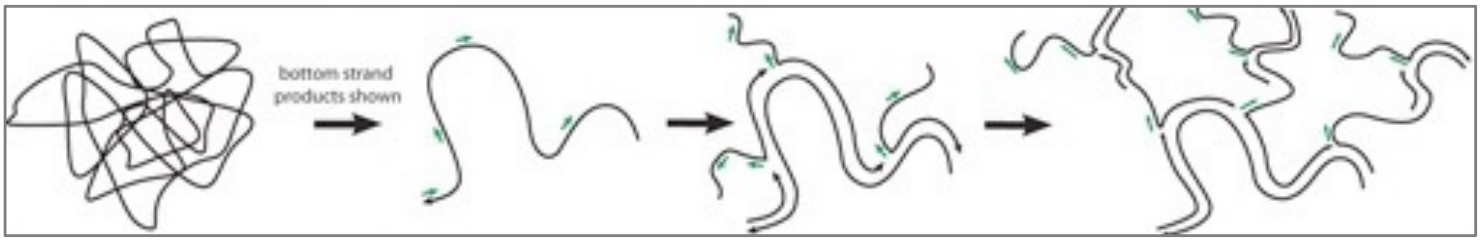


Brian Owens, Nature News 2012

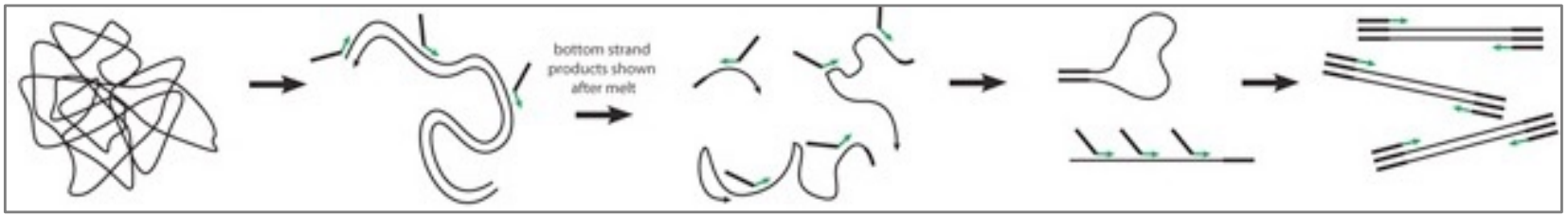
Whole Genome Amplification Techniques



DOP-PCR: Degenerate Oligonucleotide Primed PCR
Telenius et al. (1992) Genomics



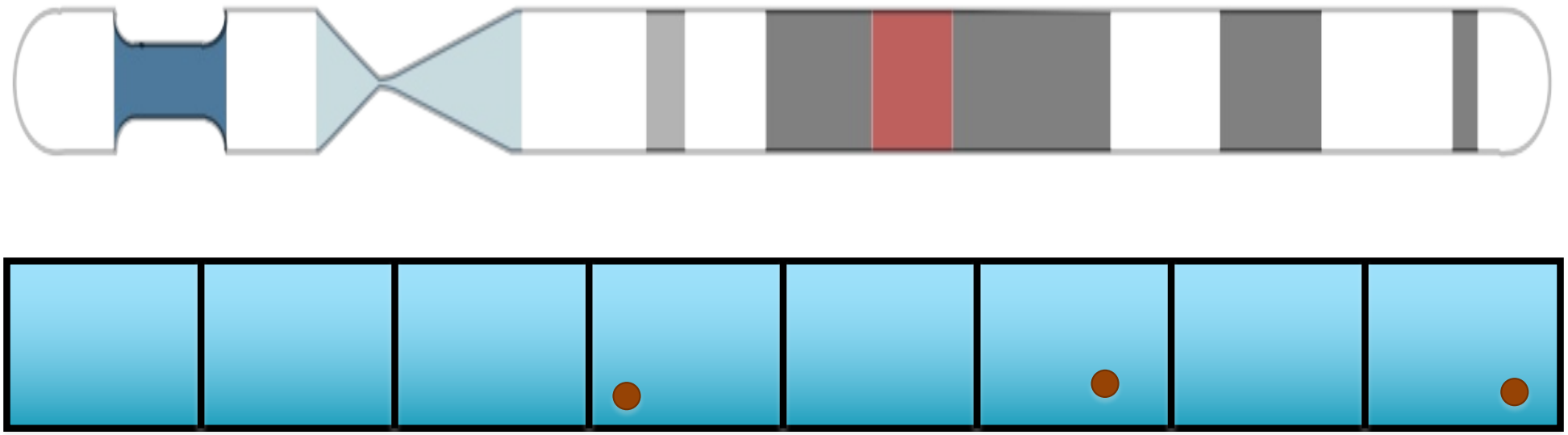
MDA: Multiple Displacement Amplification
Dean et al. (2002) PNAS



MALBAC: Multiple Annealing and Looping Based Amplification Cycles
Zong et al. (2012) Science

Paul Blainey, FEMS Microbiol Rev. 2013

1) Binning

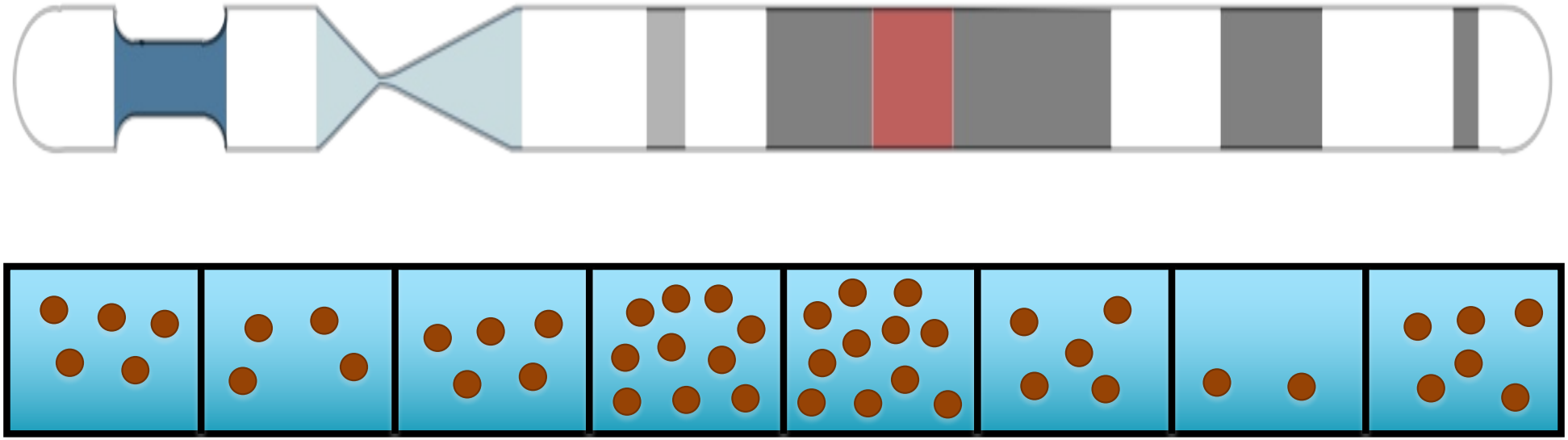


Single Cell CNV analysis

- Divide the genome into "bins" with ~50 – 100 reads / bin
- Map the reads and count reads per bin

Use uniquely mappable bases to establish bins

1) Binning



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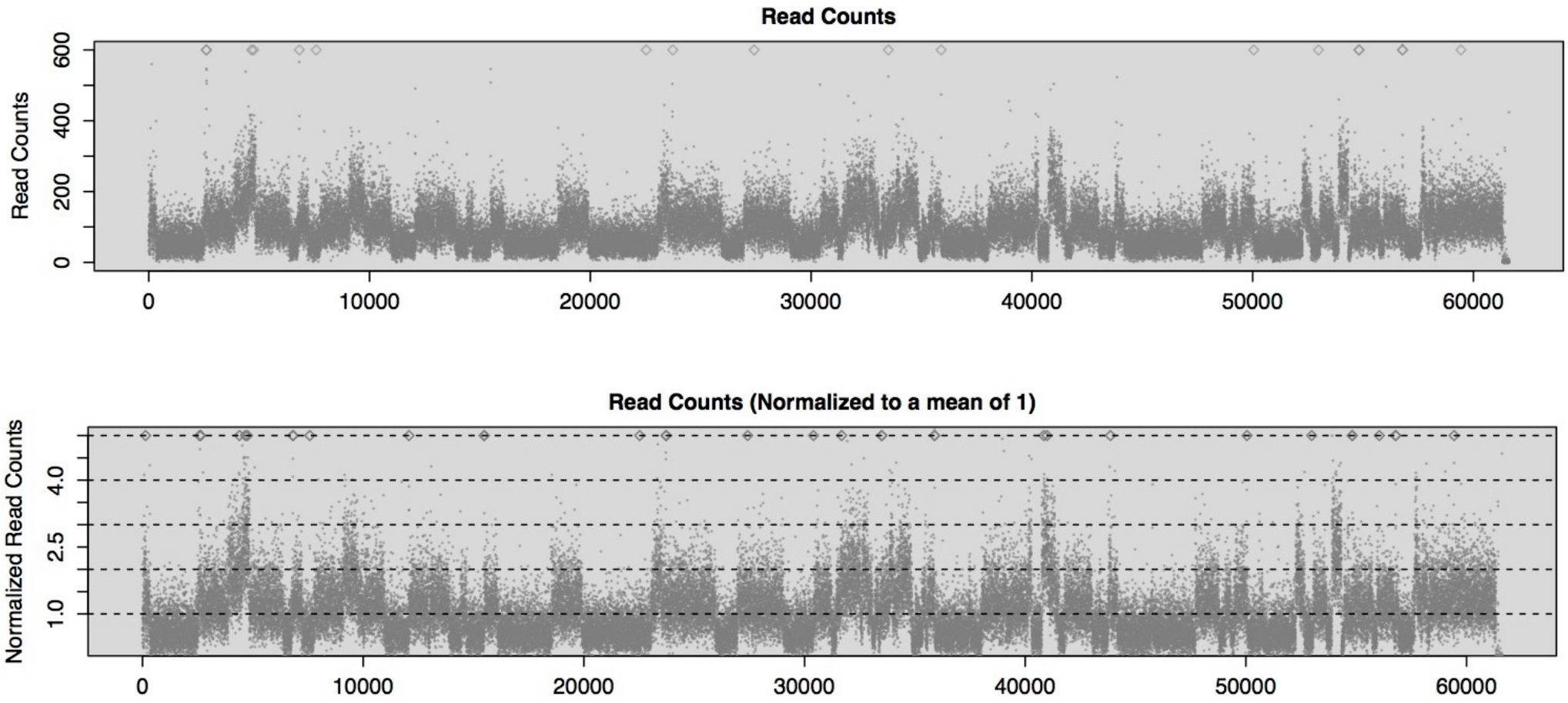
5	4	5	10	11	5	2	5

Single Cell CNV analysis

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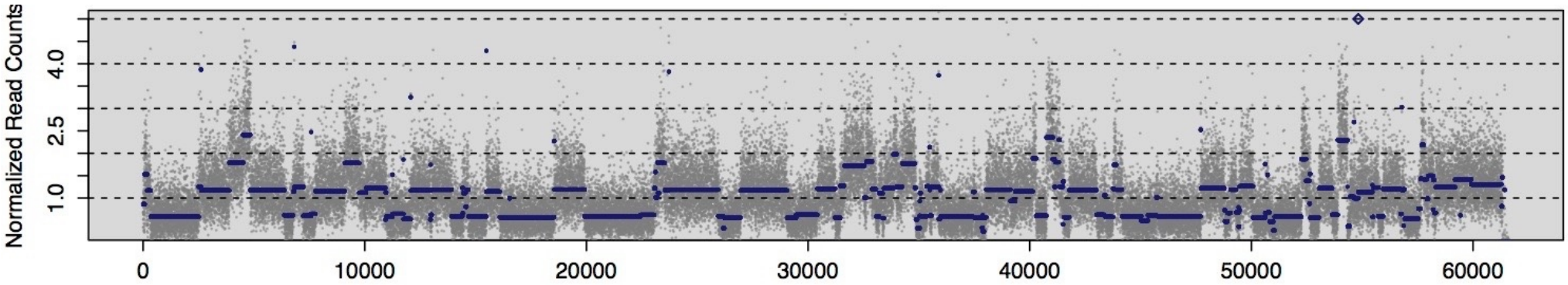
Use uniquely mappable bases to establish bins

2) Normalization

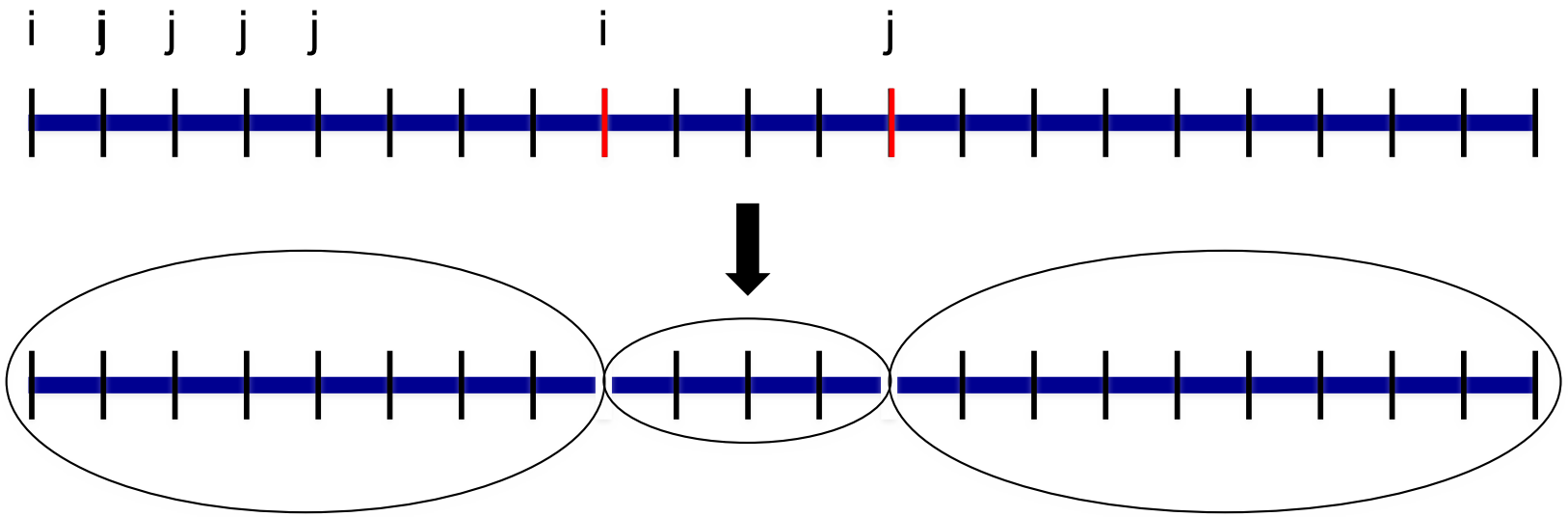


Also correct for mappability, GC content, amplification biases

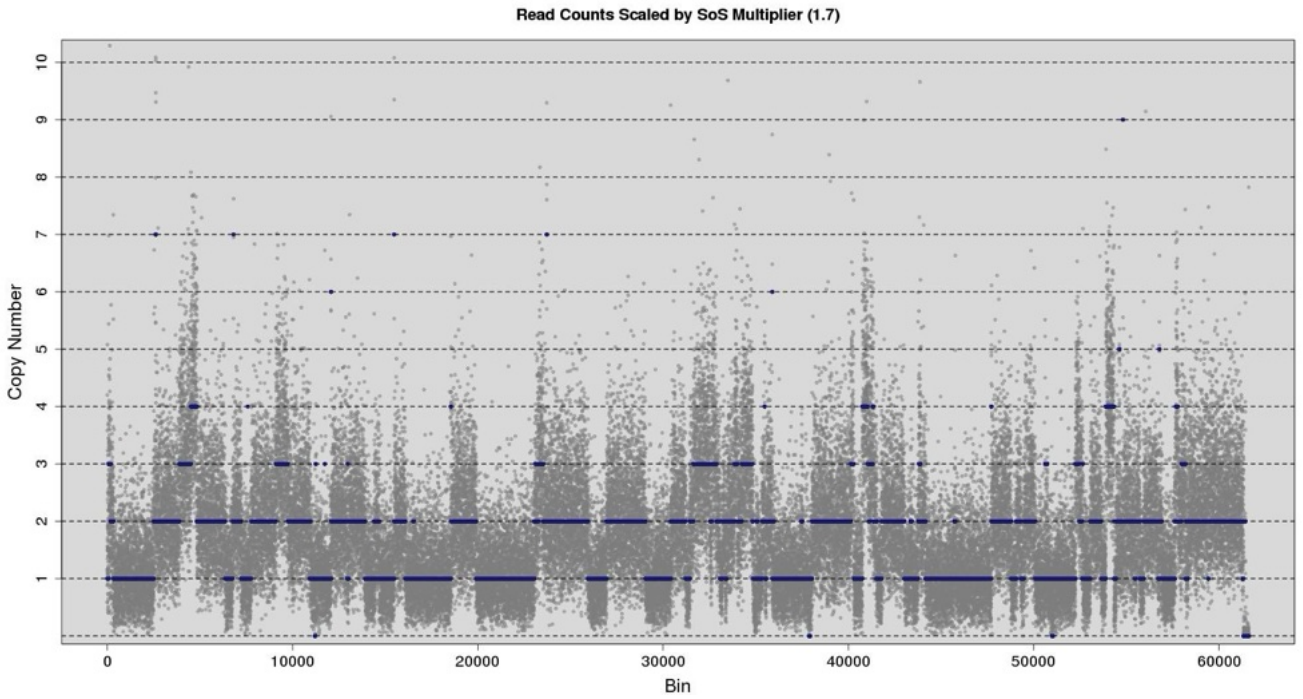
3) Segmentation



Circular Binary Segmentation (CBS)

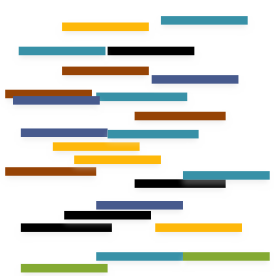


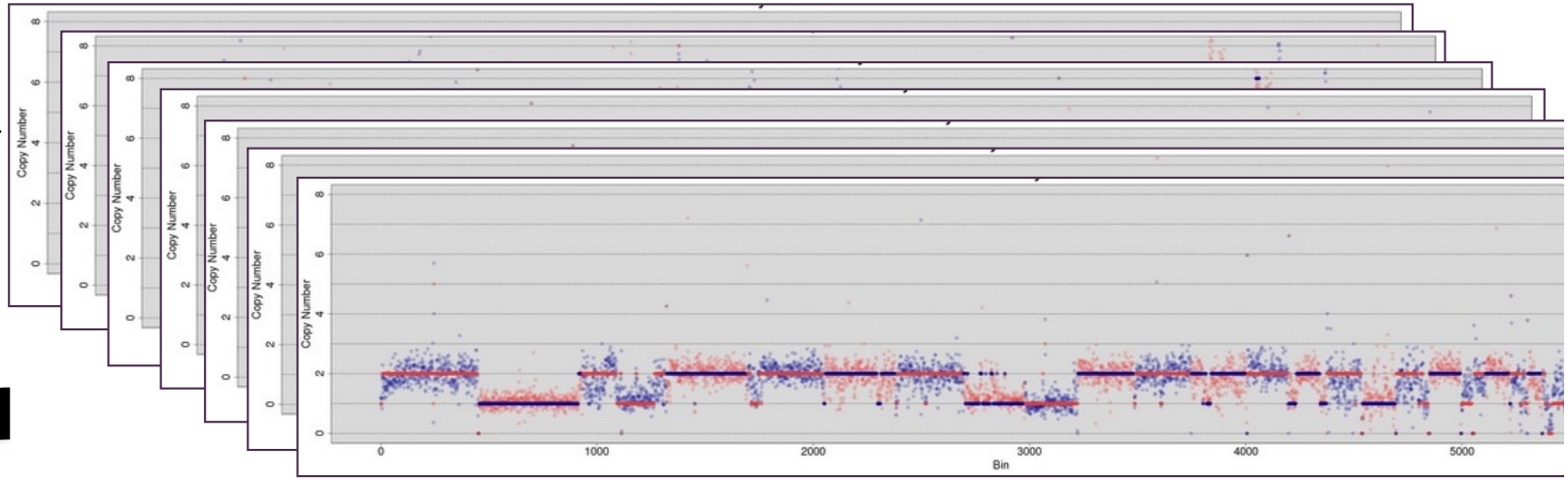
4) Estimating Copy Number

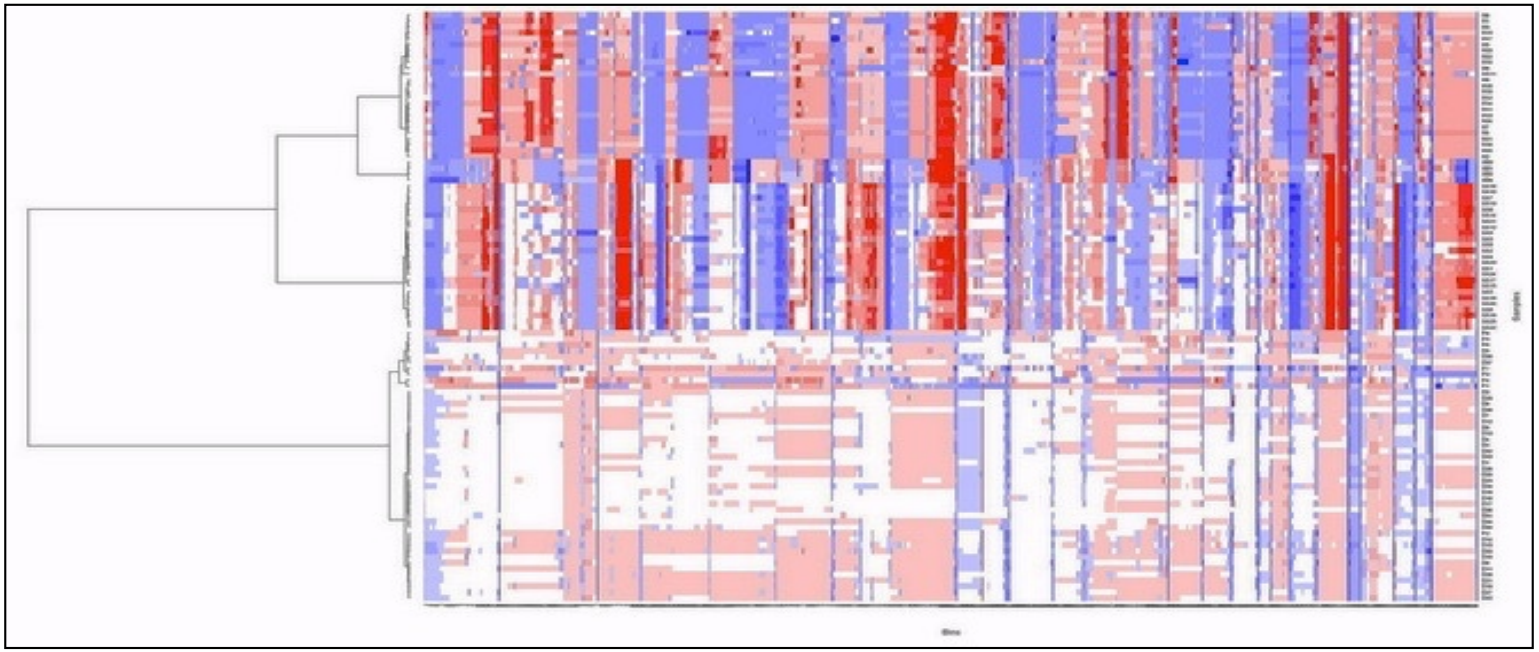


$$CN = argmin\left\{\sum_{i,j}(\hat{Y}_{i,j} - Y_{i,j})^2\right\}$$

5) Cells to Populations

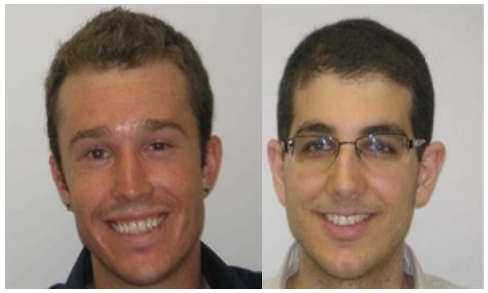


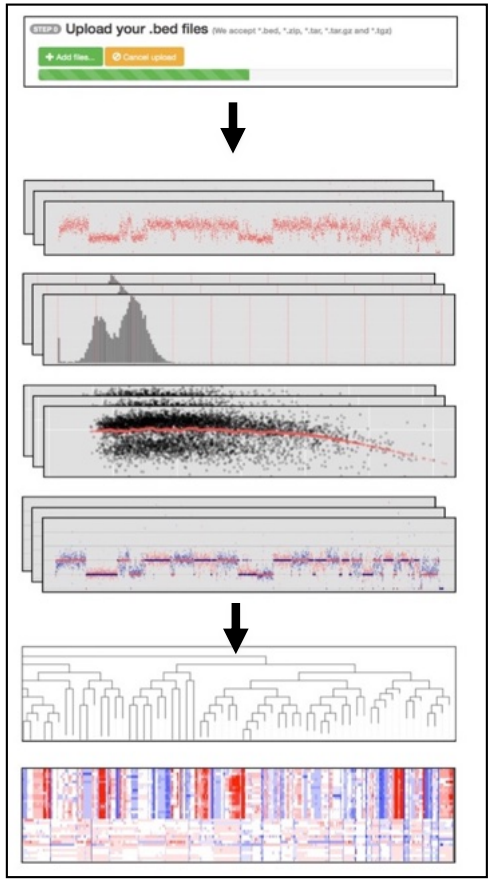




Gingko

http://qb.cshl.edu/ginkgo





Interactive Single Cell CNV analysis & clustering

- Easy-to-use, web interface, parameterized for binning, segmentation, clustering, etc
- Per cell through project-wide analysis in any species

Compare MDA, DOP-PCR, and MALBAC

- DOP-PCR shows superior resolution and consistency

Available for collaboration

- Analyzing CNVs with respect to different clinical outcomes
- Extending clustering methods, prototyping scRNA

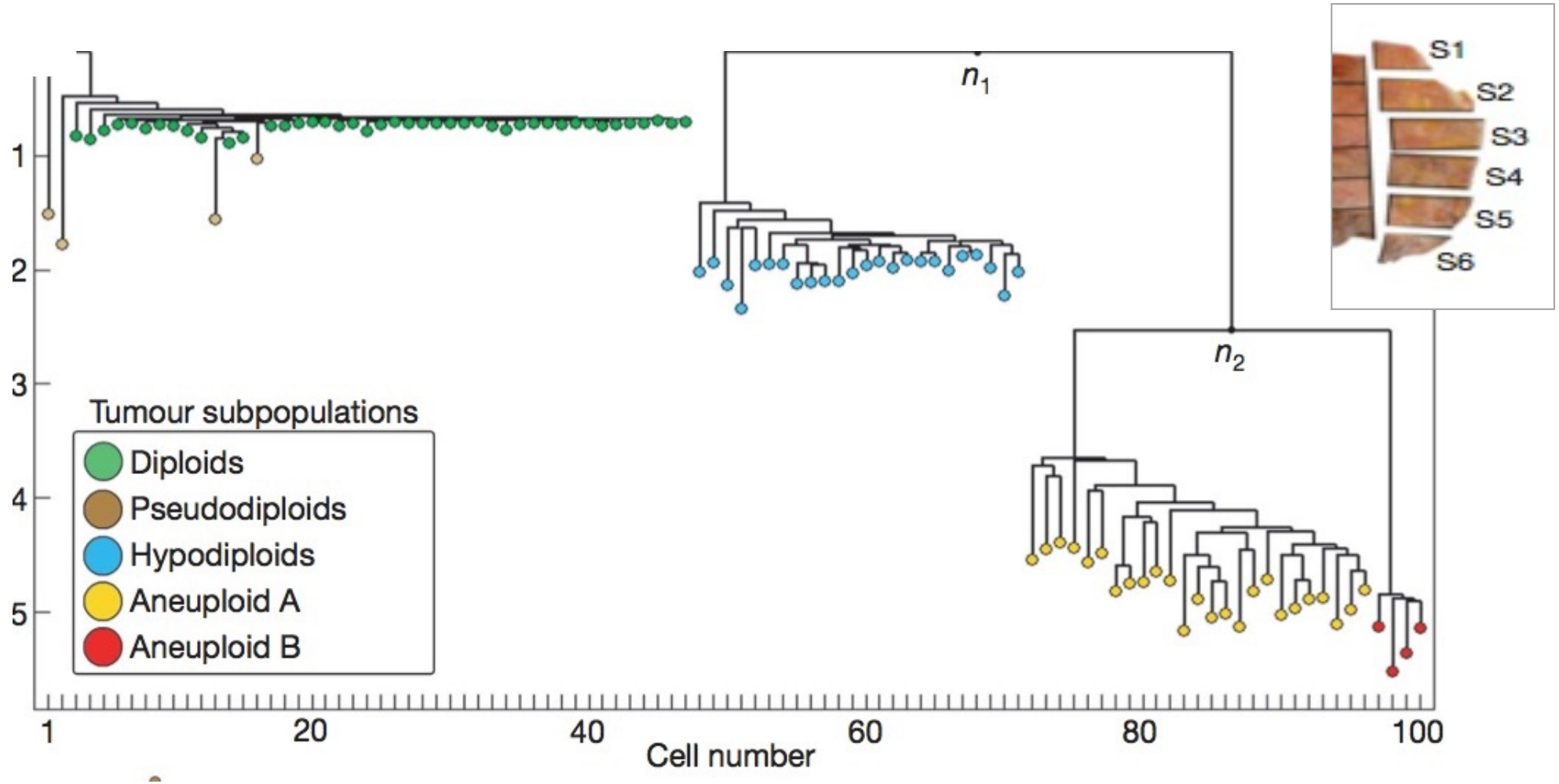
Interactive analysis and assessment of single-cell copy-number variations.
Garvin T, Aboukhalil R, Kendall J, Baslan T, Atwal GS, Hicks J, Wigler M, Schatz MC (2015)
Nature Methods doi:10.1038/nmeth.3578

LETTER

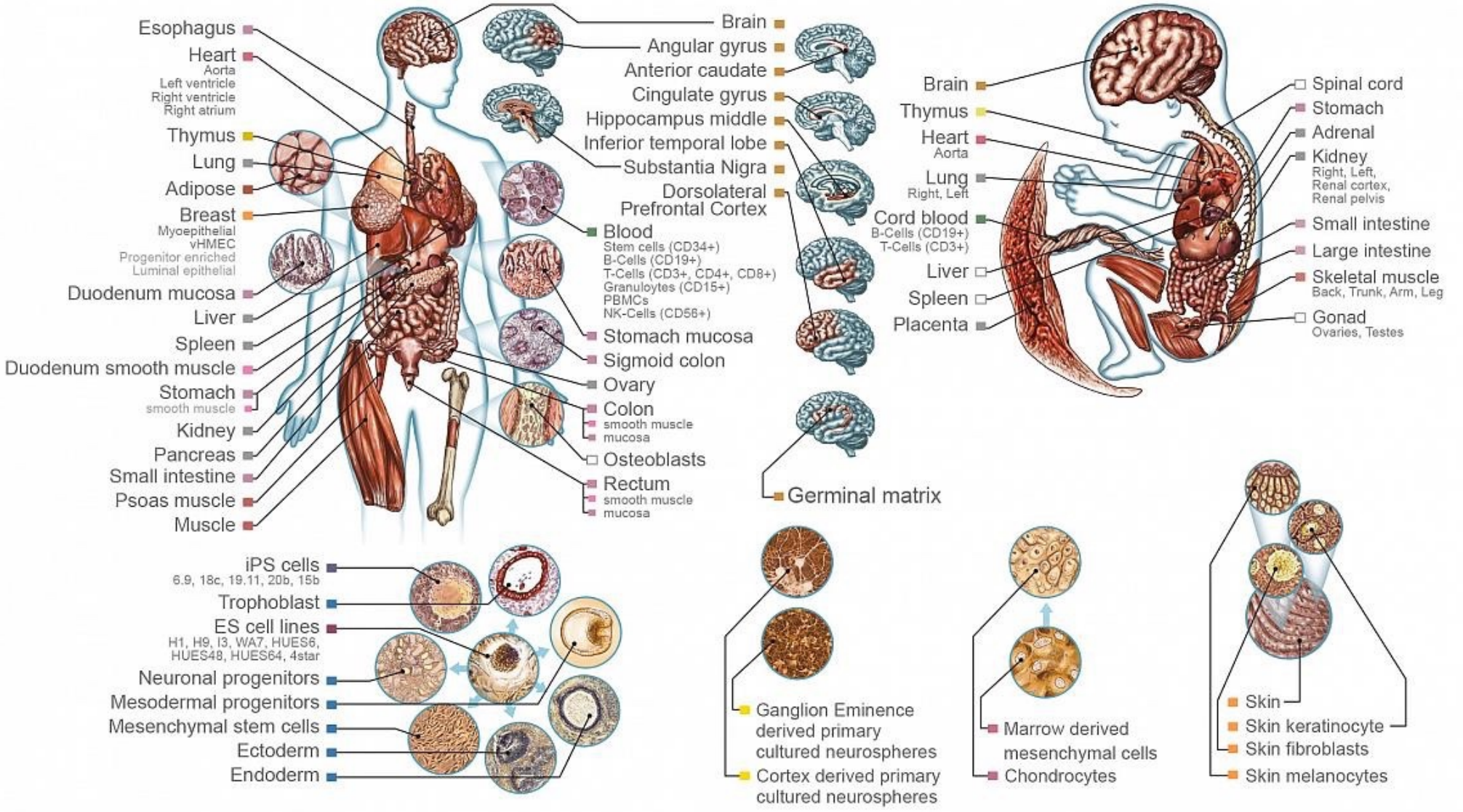
doi:10.1038/nature09807

Tumour evolution inferred by single-cell sequencing

Nicholas Navin[1,2], Jude Kendall[1], Jennifer Troge[1], Peter Andrews[1], Linda Rodgers[1], Jeanne McIndoo[1], Kerry Cook[1], Asya Stepansky[1], Dan Levy[1], Diane Esposito[1], Lakshmi Muthuswamy[3], Alex Krasnitz[1], W. Richard McCombie[1], James Hicks[1] & Michael Wigler[1]



Sources of (Cellular) Heterogeneity



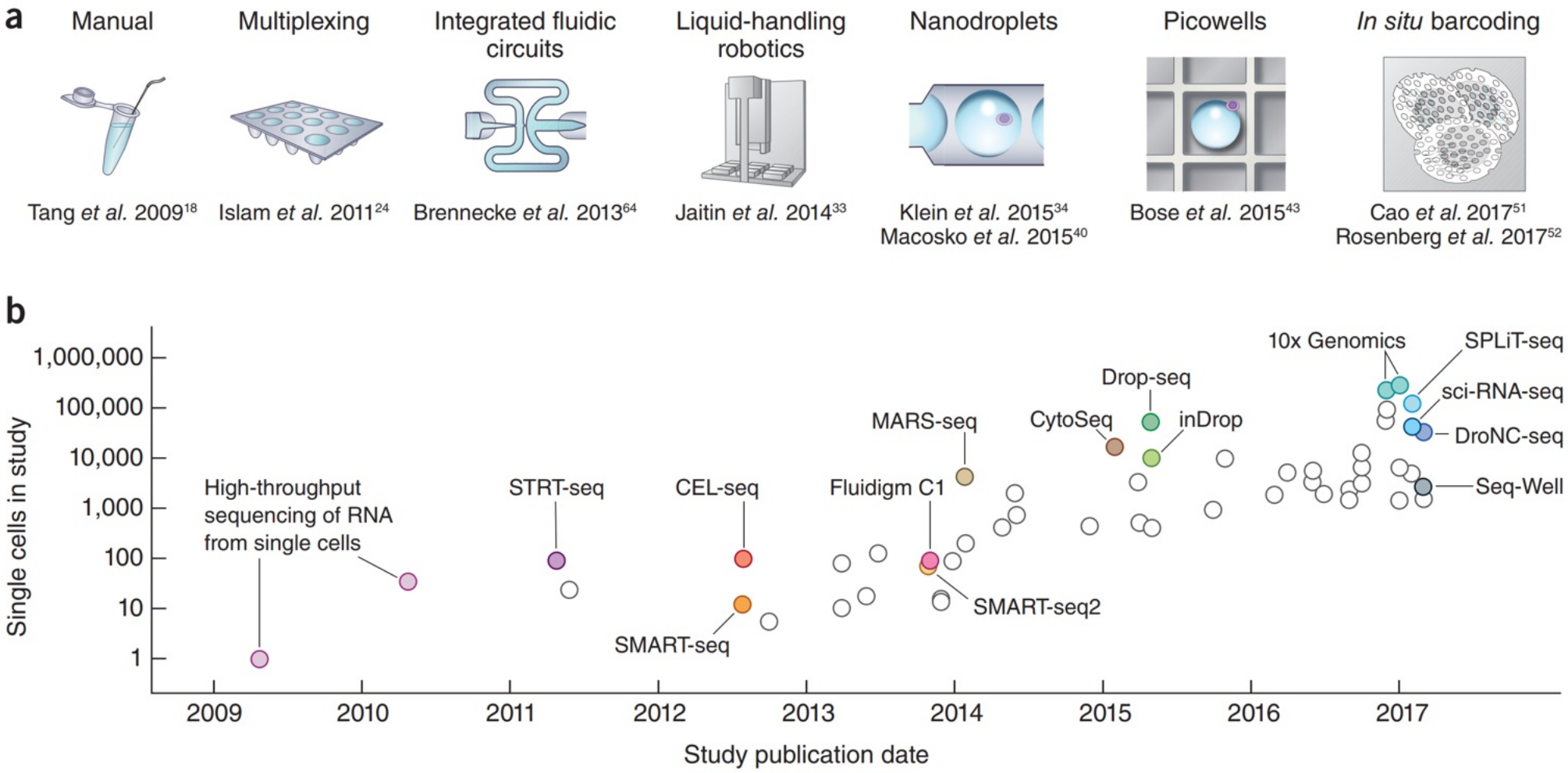
Roadmap Epigenomics Consortium

Single-cell and spatial omics

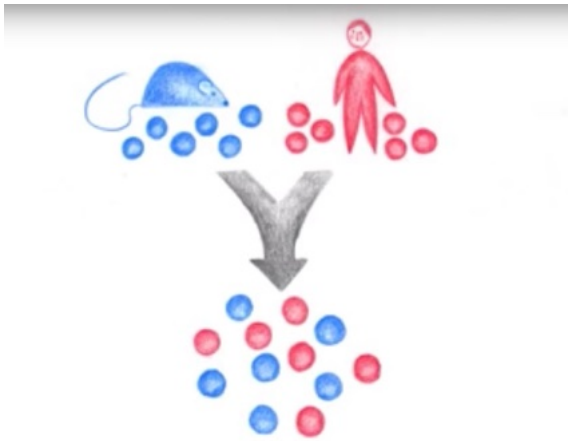
Stephanie Hicks
Department of Biomedical Engineering
Department of Biostatistics
https://stephaniehicks.com

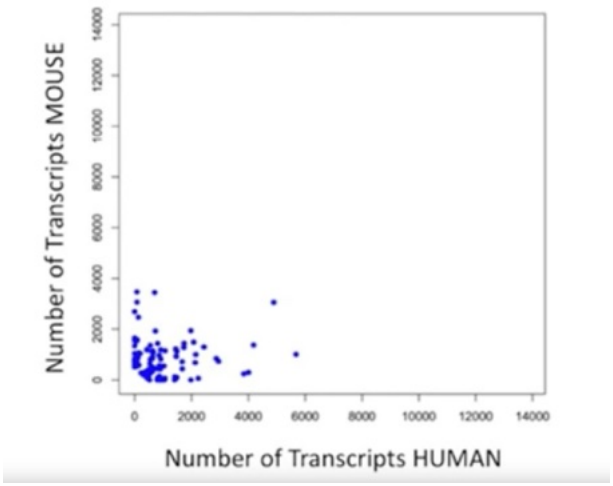
Lecture 15 -- Applied Comparative Genomics

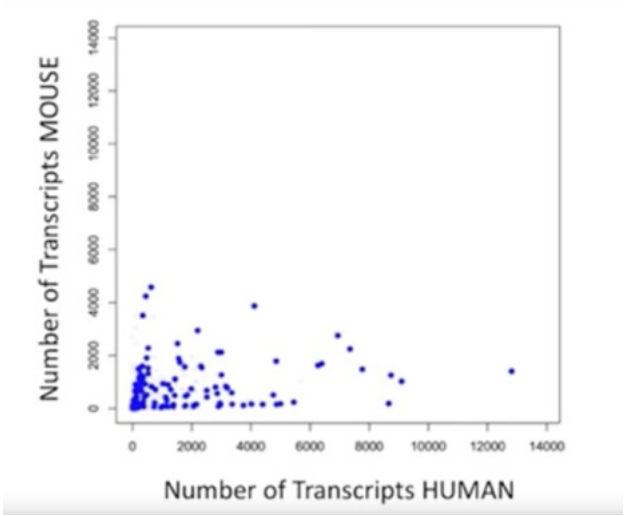
A decade of single-cell RNA-seq

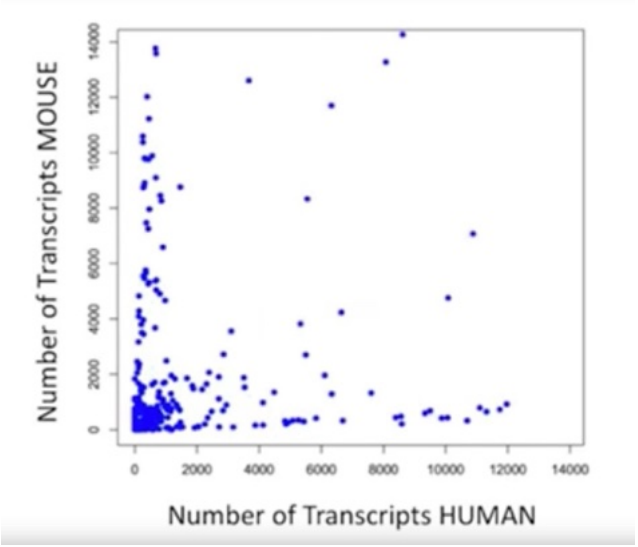


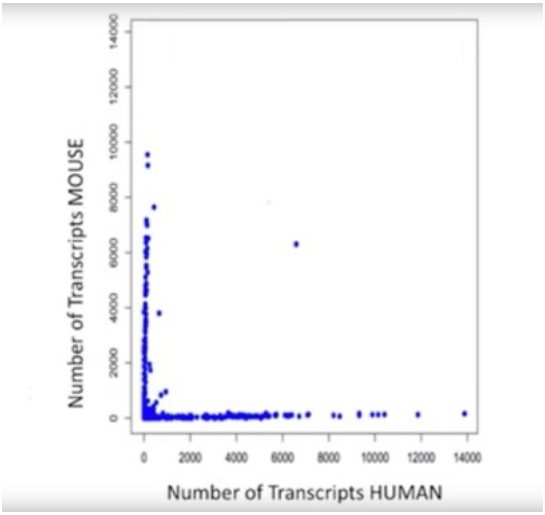
Svensson et al. 2018. DOI:10.1038/nprot.2017.149

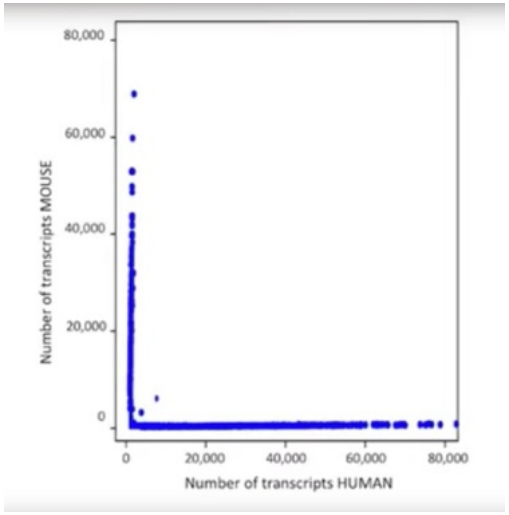












Drop-seq: Droplet barcoding of single cells
https://www.youtube.com/watch?v=vL7ptq2Dcf0

Single-cell RNA sequencing, "the bioinformatician's microscope"

— a snapshot of the underlying biology in a data matrix.

Brain cells



Biological sample

number of times **gene i** was expressed in **cell j**



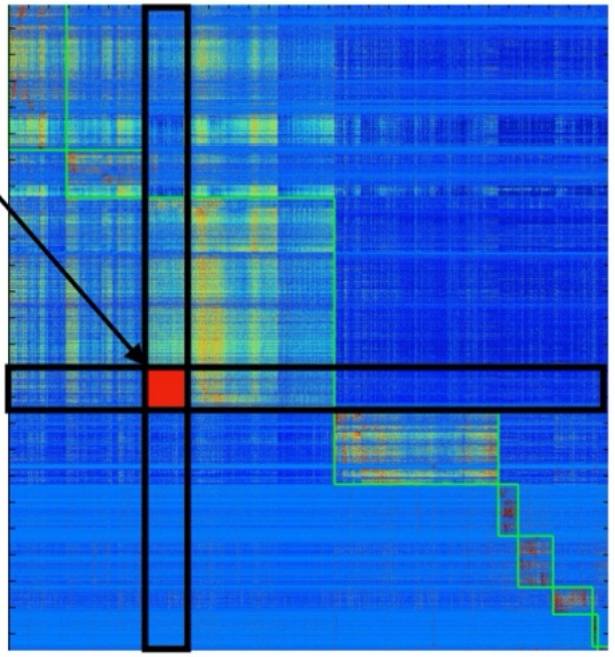
sequencing

Single Cells (samples)

cell j

Genes (features)

gene i



Gene expression matrix

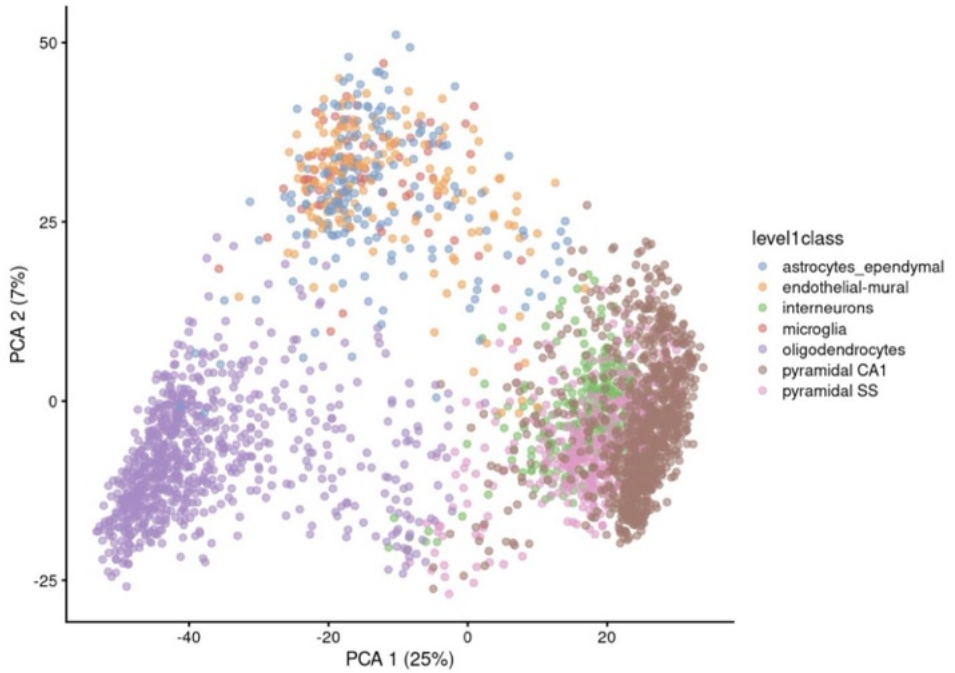
computationally explore complex biological systems

Martin Zhang

Principal Components Analysis (PCA)

Using top features selected, PCA finds orthogonal vectors (linear combinations of features in original dataset) that capture the largest amount of variation

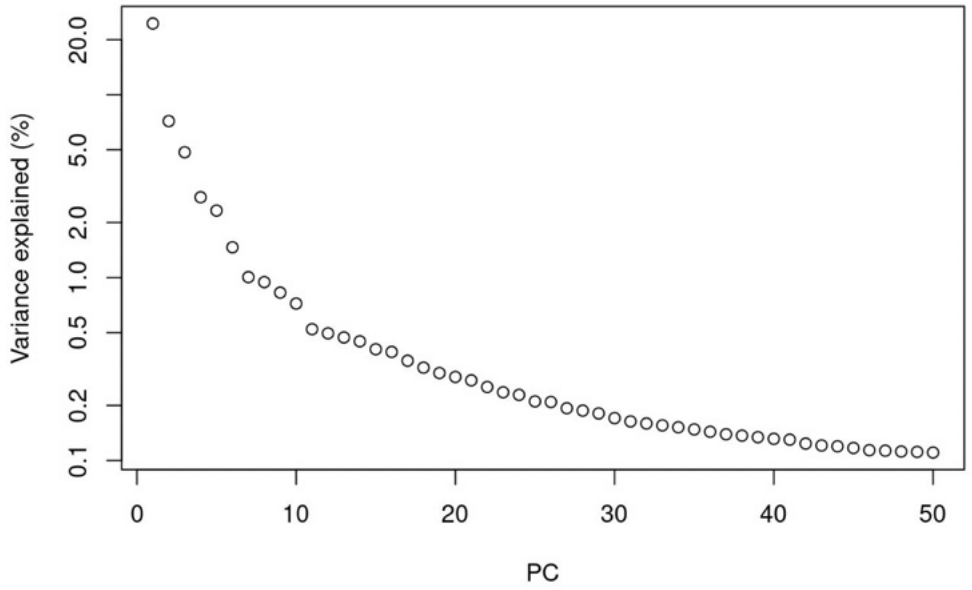
- Highly interpretable and computationally efficient
- PCs with the lowest variance are discarded to effectively reduce the dimensionality of the data without losing information
- Top PCs are typically used for downstream (e.g. clustering)



How to choose the number of PCs?

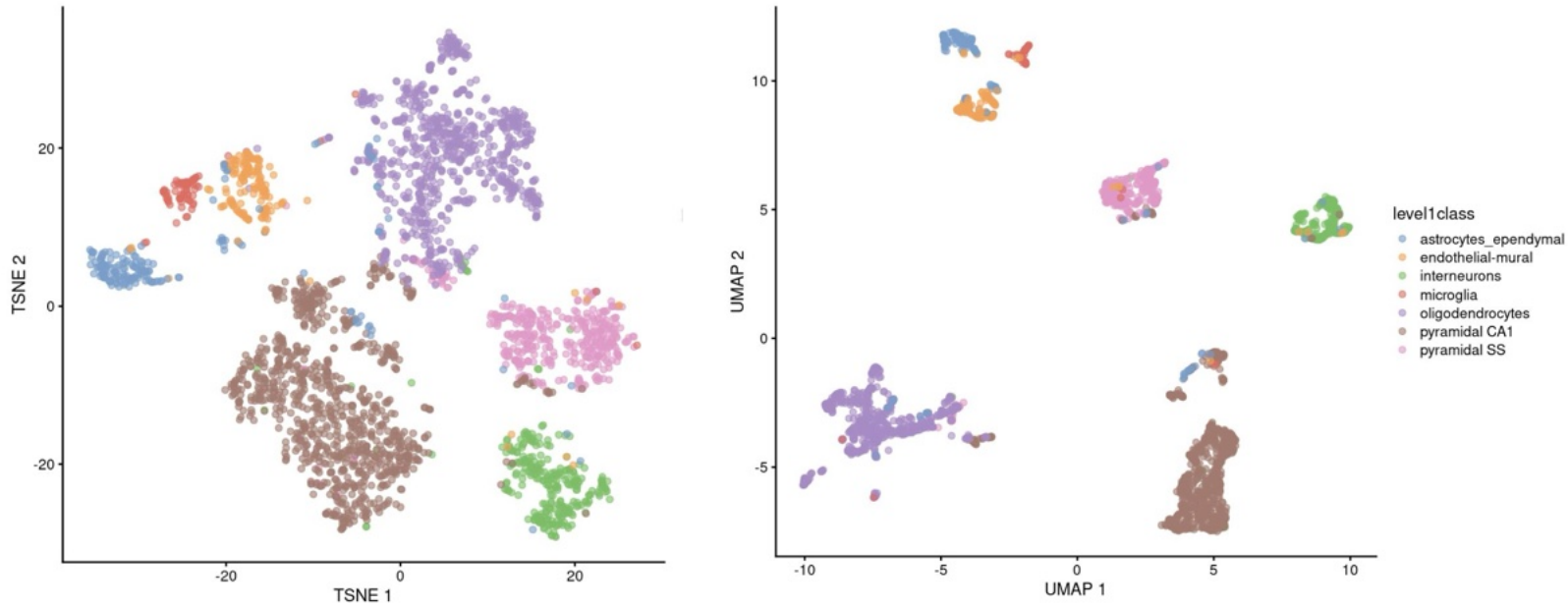
More top PCs retains more biological information, but too much introduces noise and takes longer to run methods

Often arbitrary # of PCs (e.g. 10-100) is chosen



Non-linear methods ***for data visualization***

- *t*-stochastic neighbor embedding
- Uniform manifold approximation and projection (UMAP)



Motivation for clustering

Goal: empirically define groups of cells with similar expression profiles.

Purpose: summarize complex scRNA-seq data into a digestible format for human interpretation.

This allows us to describe population heterogeneity in terms of discrete labels that are easily understood, rather than attempting to comprehend the high-dimensional manifold on which the cells truly reside.

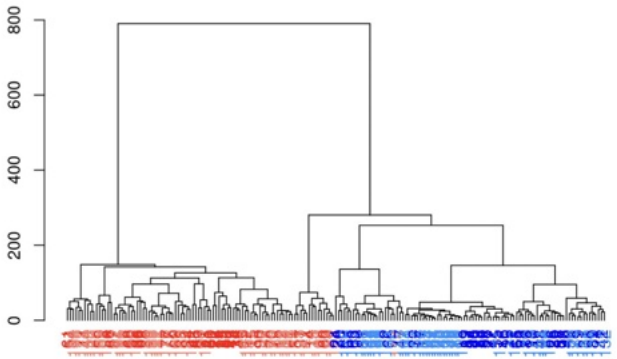
After annotation based on marker genes, the clusters can be treated as proxies for more abstract biological concepts such as cell types or states.

Hierarchical clustering

1. Calculate a pairwise cell-to-cell distance matrix
2. Arrange cells into a hierarchy based on relative similarity to each other
3. Build and cut dendrogram based on pairwise distance matrix

Pros: Visually fantastic to quantitatively capture pop relationships at various resolutions

Cons: Way too slow (*quadratic* in time for increasing # of cells); greedy algorithm (i.e. likely a suboptimal partition)

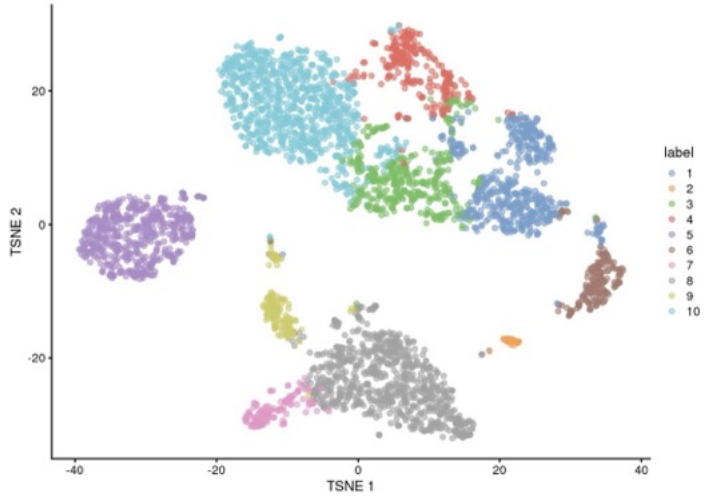


k-means clustering

1. Assignment: Given centroids, assign a cell to the cluster with the closest centroid
2. Update: Compute new centroid profiles for each cluster

Pros: Super simple and easy algorithm to understand; scalable to thousands of cells

Cons: Assumption that the data have spherical variance



Graph-based clustering

Idea:

1. Build a graph structure where each node is a cell that is connected to its neighbs. Edges are weighted based on the similarity between the cells
2. Apply algorithms to identify “communities” of cells that are more connected to cells in the same community than they are to cells of different communities. Each community represents a cluster that we can use for downstream interpretation.

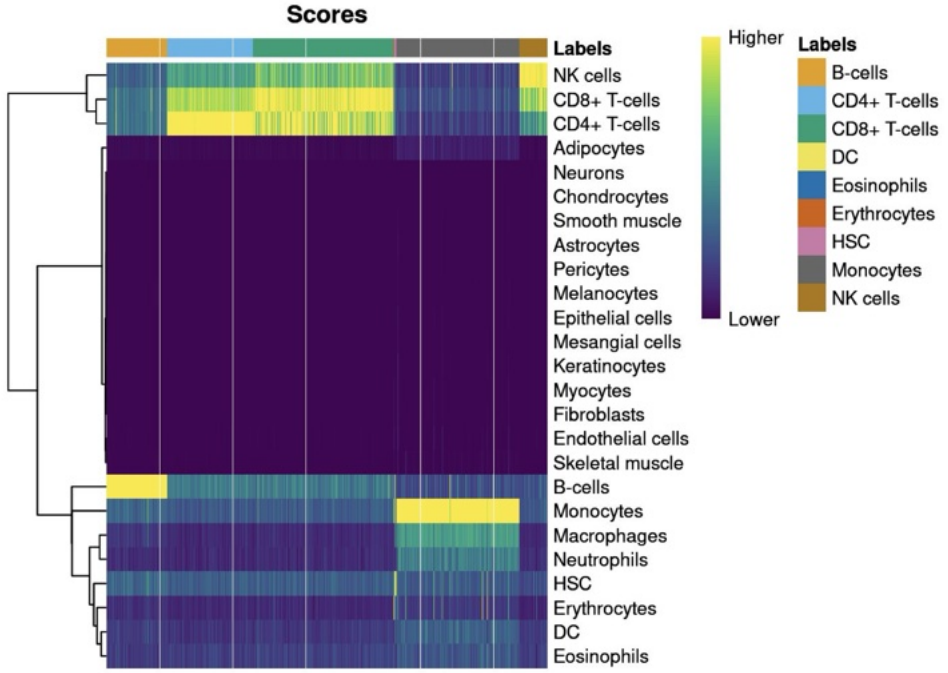
Pros: (1) Super scalable (ie only needs *k*-NN search log-linear time vs quadratic) in # of cells, and (2) Graph avoids making strong assumptions about the shape of the clusters

Cons: after graph, no information is retained about relationships beyond cells

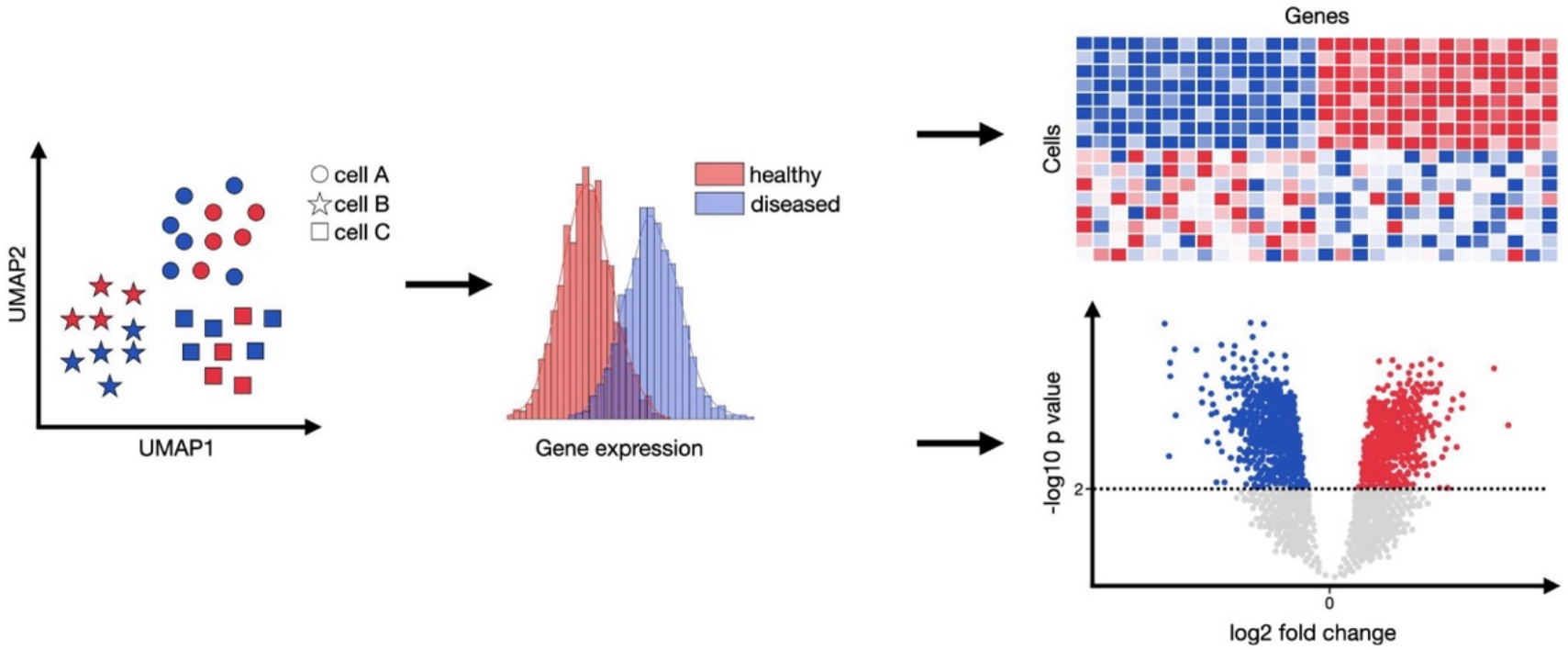
Cell type annotation / labeling

Assign cell labels using reference datasets

- Correlation to training data
- Or simply plot gene markers



Differential expression (DE) analysis



https://www.sc-best-practices.org/conditions/differential_gene_expression.html

Differential expression (DE) analysis

Common DE approaches:

- Cell level:
  - *t*-test (comparing the differences in mean expression between two groups)
  - Wilcoxon Rank Sum Test (basically same as *t*-test, but nonparametric)
  - Differences in distributions (link)
  - Generalized mixed effect models such as MAST (link) or glmmTMB (link)
- Sample/donor-level
  - Aggregate cell counts to create “pseudobulks” and then analysed with methods originally designed for bulk expression samples such as edgeR (link) or DEseq2 (link)

How to choose? My general two cents...

- Pseudobulk approaches tend to be more robust as single-cell data is noisy and sparse!

Korthauer *et al. Genome Biology* (2019) 20:118
https://doi.org/10.1186/s13059-019-1716-1

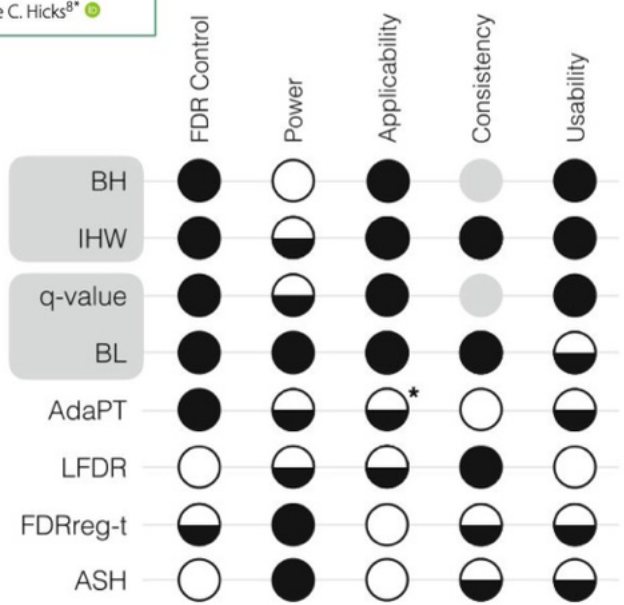
Genome Biology

RESEARCH Open Access

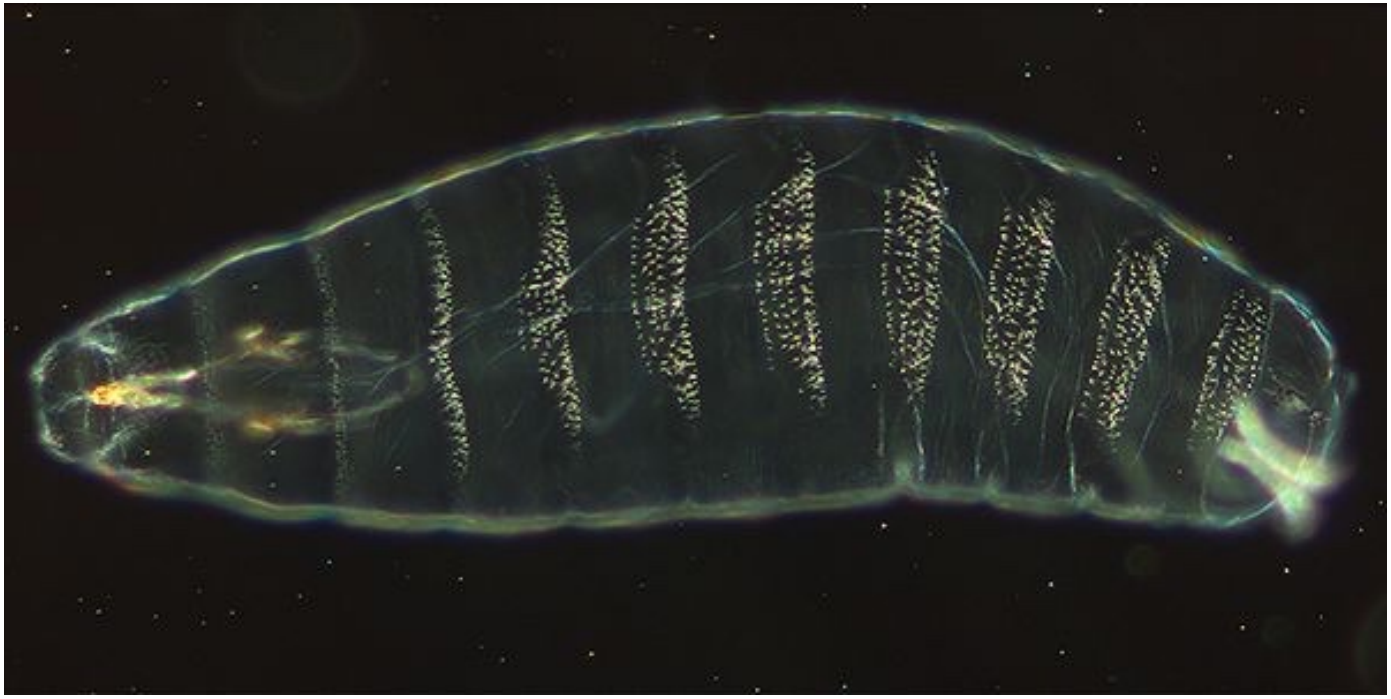
A practical guide to methods controlling false discoveries in computational biology

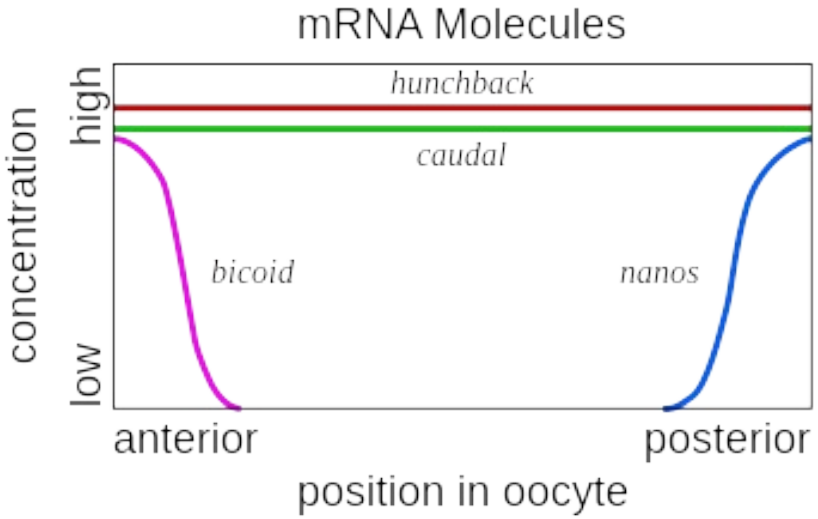
Keegan Korthauer[1,2†], Patrick K. Kimes[1,2†], Claire Duvallet[3,4†], Alejandro Reyes[1,2†], Ayshwarya Subramanian[5†], Mingxiang Teng[6], Chinmay Shukla[7], Eric J. Alm[3,4,5] and Stephanie C. Hicks[8*]

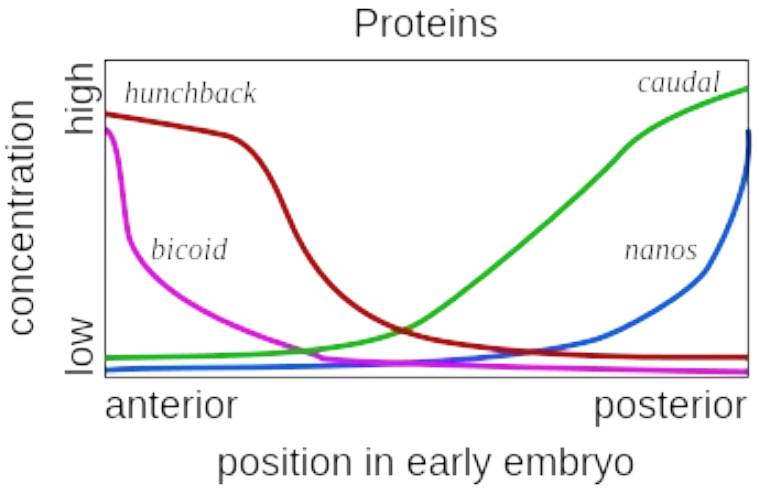
	Input	Assumptions	Output	R package
BH	*p*-values	exchangeability	adjusted *p*-values	`stats`
IHW	(1) *p*-values (2) independent & informative covariate	exchangeability within covariate groups	adjusted *p*-values	`ihw`
q-value	*p*-values	exchangeability	*q*-values	`qvalue`
BL	(1) *p*-values (2) independent & informative covariate	exchangeability conditional on covariate(s)	adjusted *p*-values	`swfdr`
AdaPT			*q*-values	`adaptMT`
LFDR		exchangeability within covariate groups	adjusted *p*-values	none
FDRreg	(1) *z*-scores (2) independent & informative covariate	exchangeability conditional on covariate(s); normal test statistics	Bayesian FDRs	`FDRreg`
ASH	(1) effect sizes (2) standard errors of (1)	effects are unimodal; test statistics have normal or *t* mixture components	*q*-values	`ash`



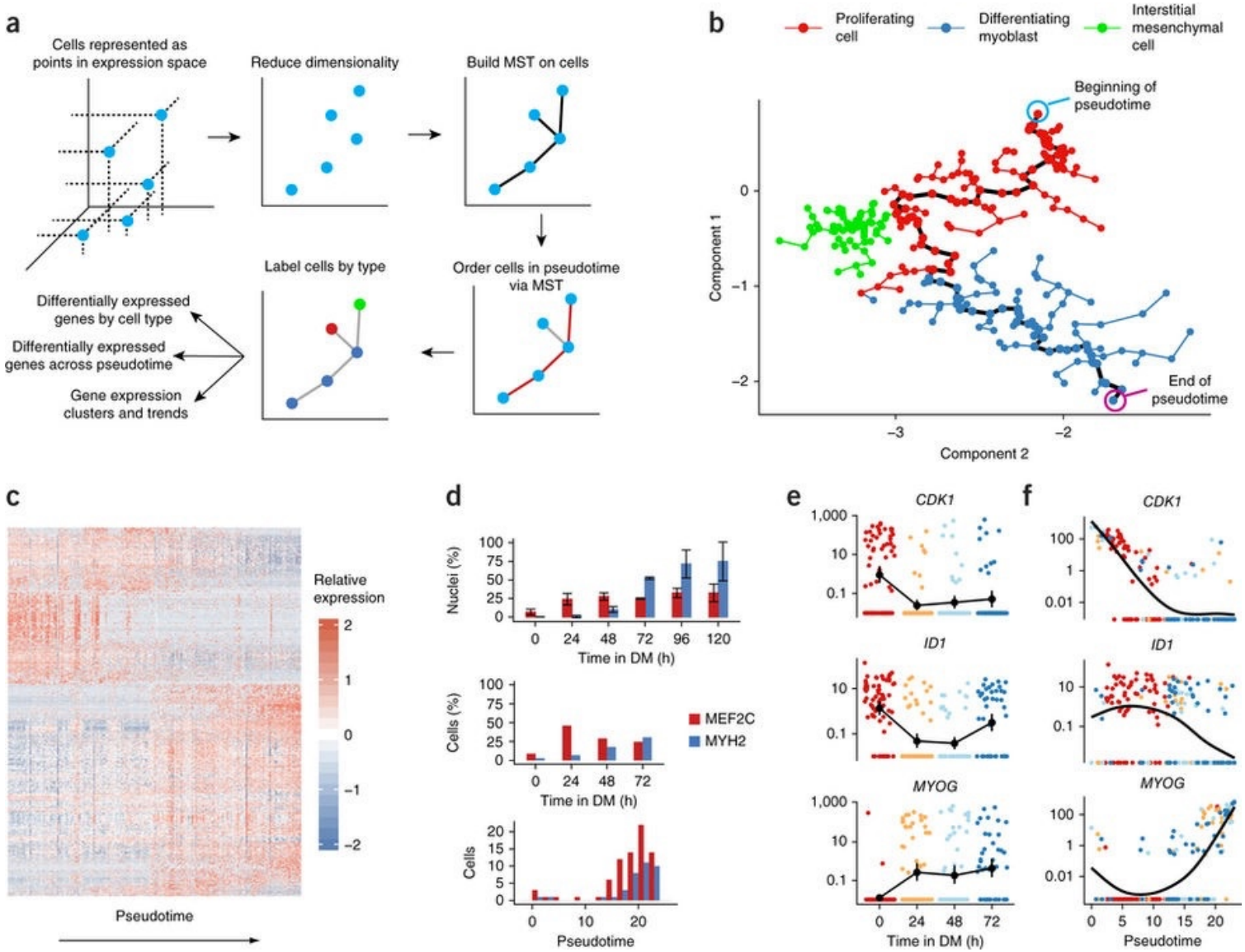
https://genomebiology.biomedcentral.com/articles/10.1186/s13059-019-1716-1







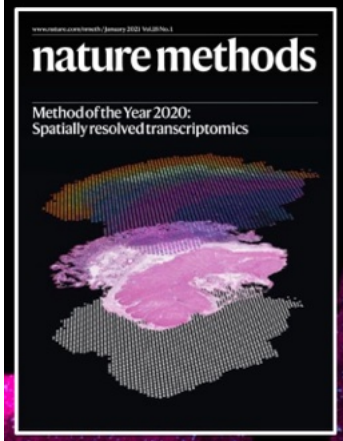
https://en.wikipedia.org/wiki/Drosophila_embryogenesis



The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells ("Monocle")

Trapnell et al (2014) Nature Biotechnology. doi:10.1038/nbt.2859

Spatially-resolved transcriptomics



High-throughput spatially resolved transcriptomics

Can view as extension of single-cell RNA sequencing with added spatial information

Example: cell populations within the brain



bulk RNA-seq

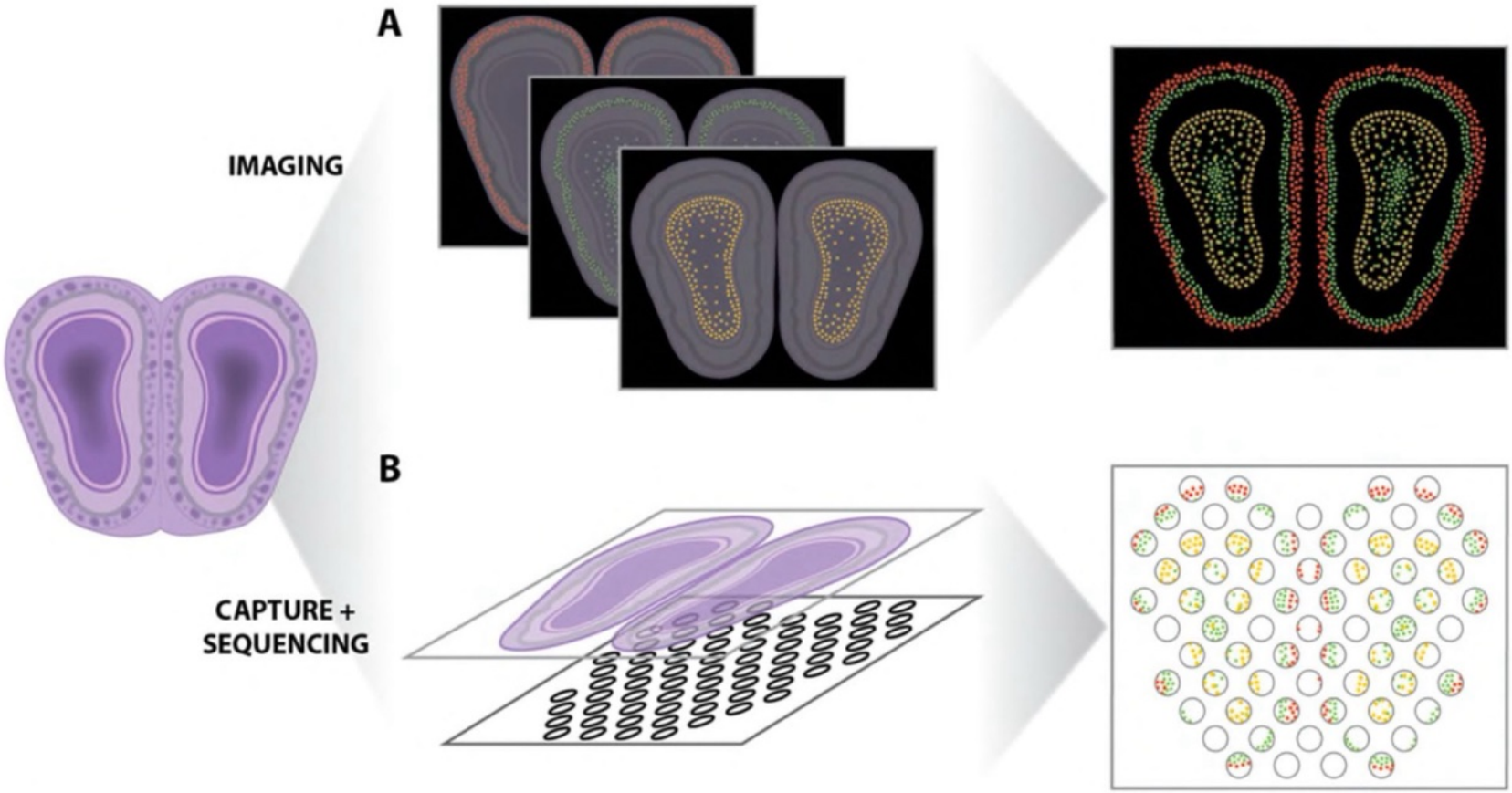


single-cell RNA-seq



spatially-resolved transcriptomics

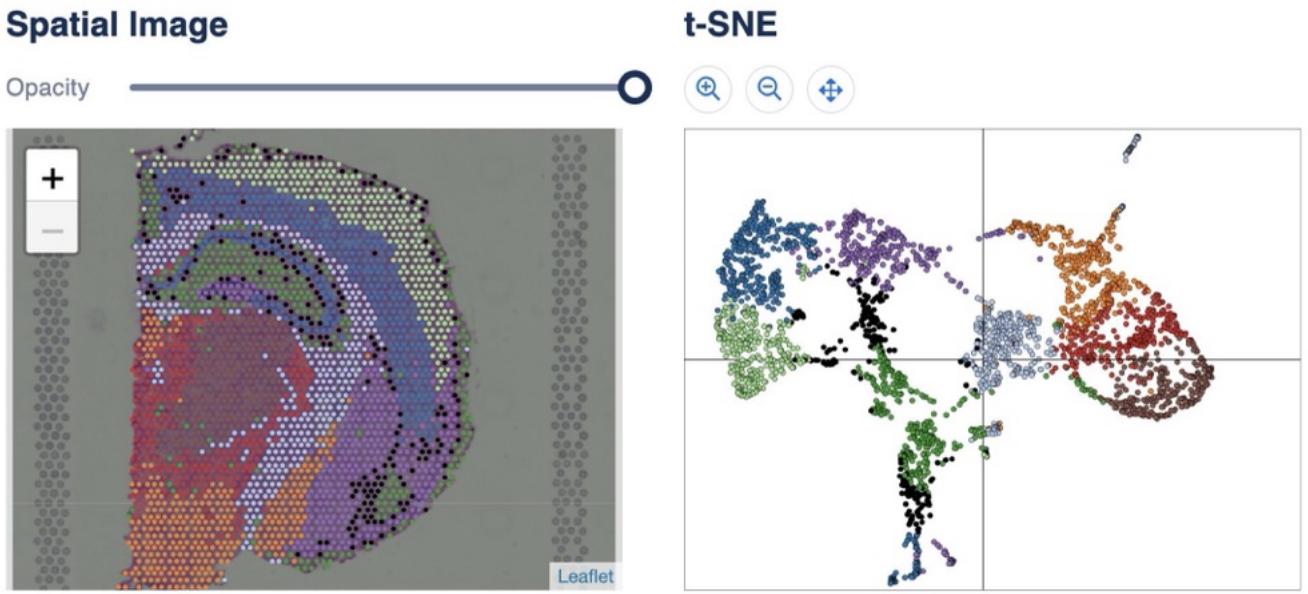
Image credit: @BoXia7



Atta and Fan (2021). *Nature Communications*

Spatially-resolved transcriptomics

Example: 10x Genomics Visium platform

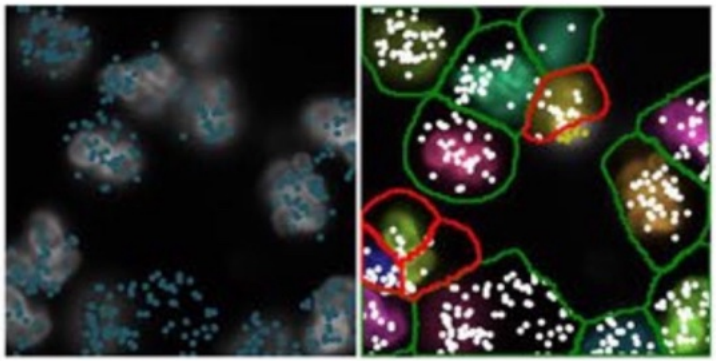


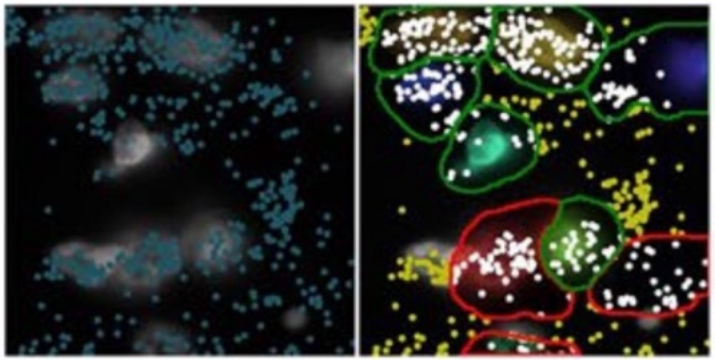
https://www.10xgenomics.com/spatial-transcriptomics

https://genomebiology.biomedcentral.com/articles/10.1186/s13059-023-03054-0

Preprocessing steps: cell / nuclei segmentation





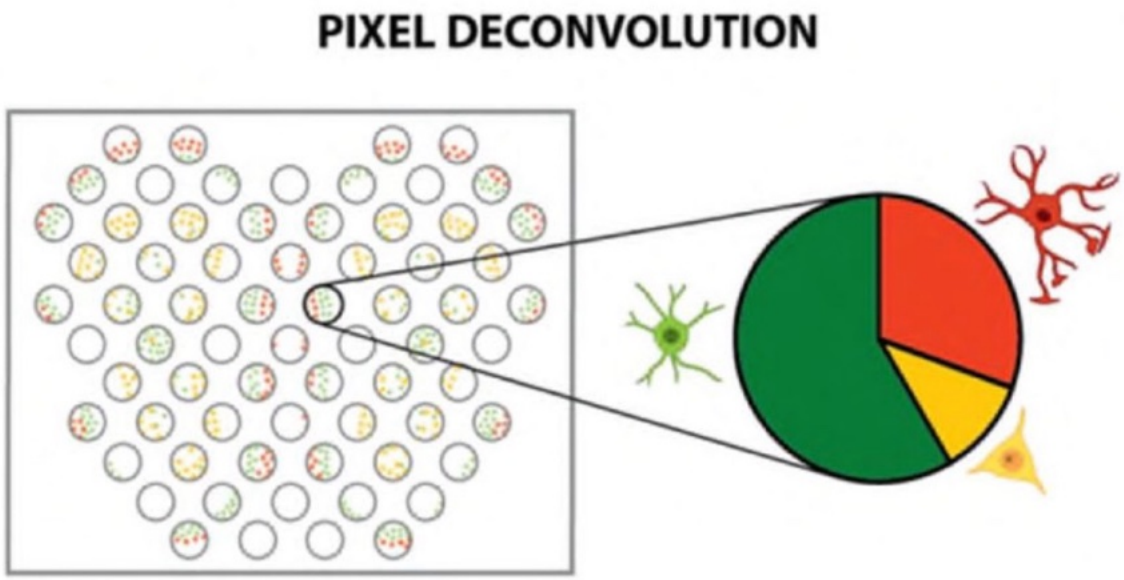


Broad approaches:

1. **Designed for biomedical images** (e.g. Watershed, U-Net, Deepcell, and Cellpose) -- often the image doesn't contain enough info to segment cell boundary (mostly stains for just nucleus)
2. **Uses spatial locations of RNA reads** to infer cell boundaries (e.g. Baysor) -- often ignores the image or makes unrealistic assumptions
3. **Integrates both** images and spatial locations of RNA reads

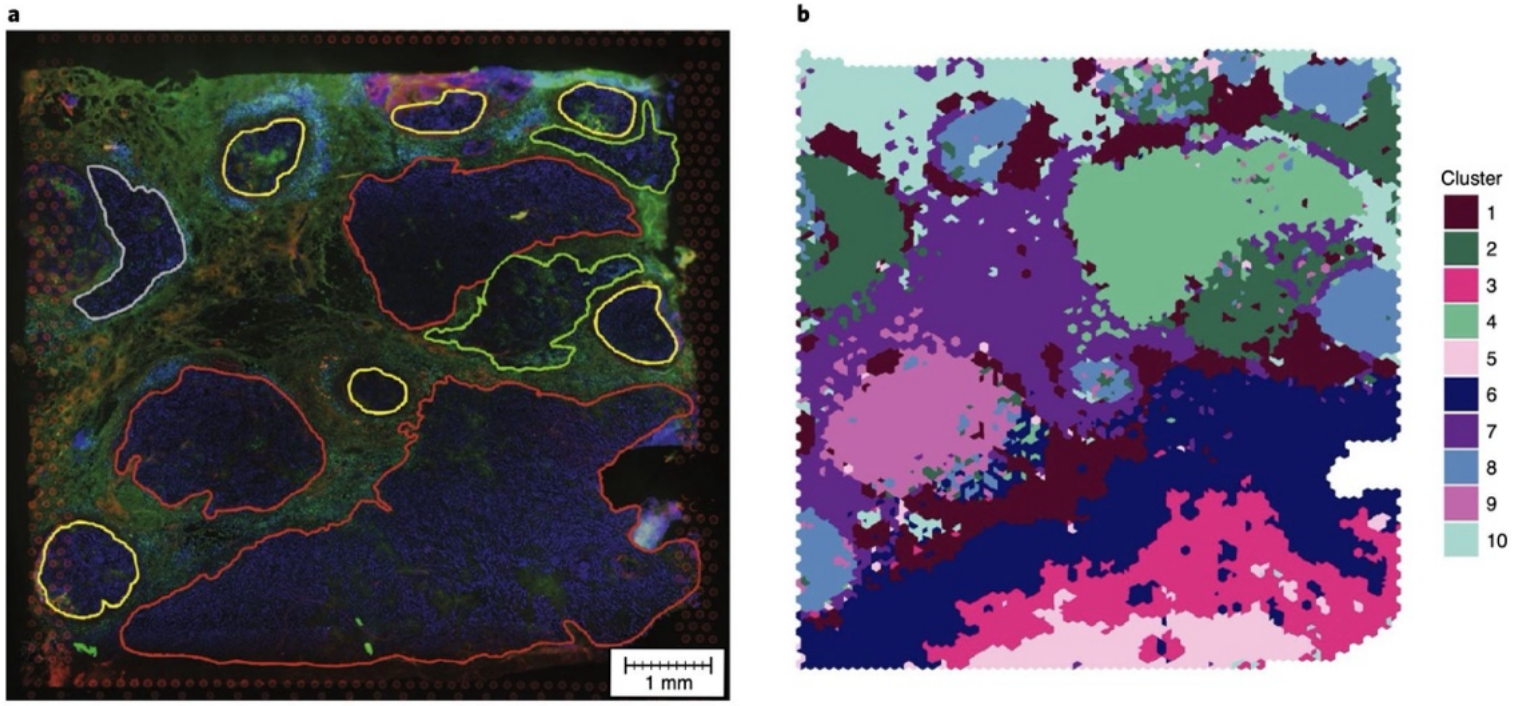
Types of questions?

Deconvolve multi-cellular pixel-resolution data to determine pixel cell-type composition at each observed spatial location (sometimes called “deconvolution”)



Atta and Fan (2021)

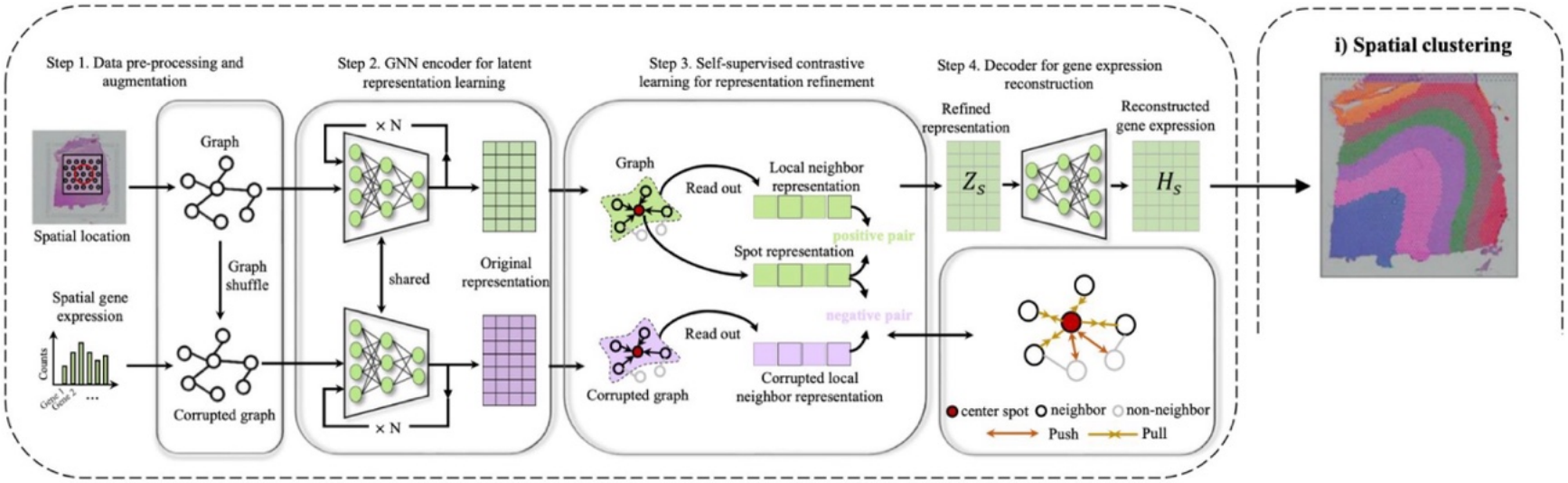
Spatial clustering with BayesSpace



Fully Bayesian statistical approach with a Markov random field

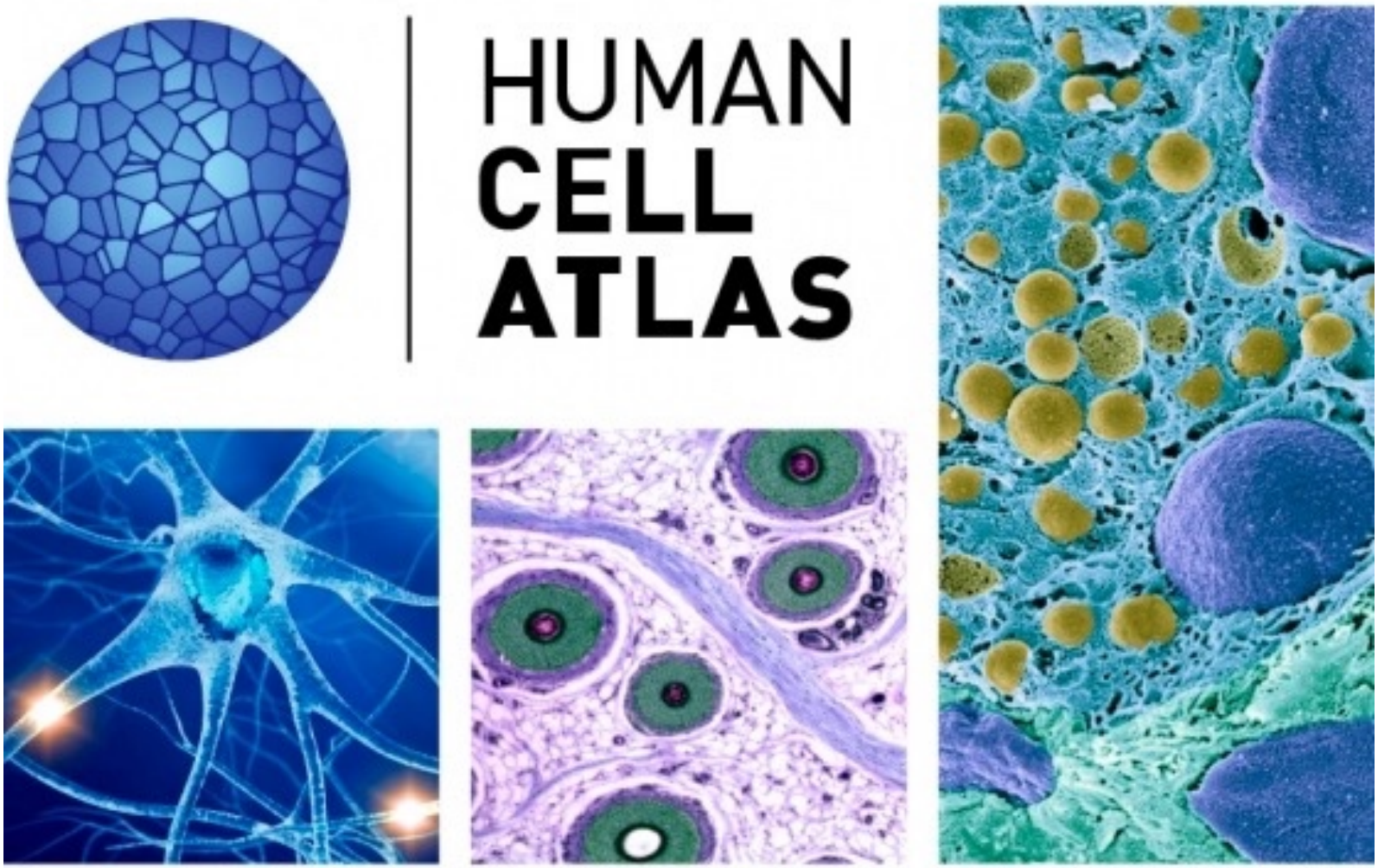
https://www.nature.com/articles/s41587-021-00935-2

Spatial clustering with GraphST



GraphST = graph-based convolutional autoencoder (and refines latent embeddings with contrastive self-supervised learning)

https://www.nature.com/articles/s41592-024-02215-8



https://www.humancellatlas.org/

scSummary

Single cell analysis is a powerful tool to study heterogeneous tissues

- Overcomes fundamental problems that can arise when averaging
- scRNA analysis used to identify novel cell types, understand the progression from one cell type to another across development or disease
- Many other sc-assays in development, expect 1000s to 1Ms of cells in essentially any assay

Major challenges

- Very sparse amplification and few reads per cell
- Find large CNVs, identify major cell types; hard to find small variants or perform differential expression
- Allelic-dropout and unbalanced amplification hides or distorts information
- Use statistical approaches to smooth results based on prior information or other cells from the same cell type

- Need new ways to process and analyze millions of cells at a time